SYLLABI

U.G. Courses offered by the Department of COMPUTER ENGINEERING

REVISED WITH EFFECT FROM 2017-18

DEPARTMENT OF COMPUTER ENGINEERING

ALIGARH MUSLIM UNIVERSITY, ALIGARH

ABOUT THE DEPARTMENT

The Department of Computer Engineering was created in the Faculty of Engineering in the year 1991. The Department is currently running an eight Semester B. Tech. programme in Computer Engineering and a four semester M. Tech programme in Computer Science & Engineering with specialization in Software Engineering. The Department also offers Ph.D. Programme in emerging areas of Computer Engineering.

The Department has the following well equipped laboratories: -

- Software Laboratory
- Hardware Laboratory
- Advanced Computing Laboratory
- Networking Laboratory
- PG Laboratory
- Basic Computing Laboratory

All Laboratories, Staff Rooms and office of the Department are fully networked and have access to high speed University Internet facility.

The Department has highly qualified, motivated and dedicated teaching faculty with education from reputed institutes. They are actively involved in Research and Development activities in the emerging areas of Computer Engineering. The prominent areas of research in the Department are - Computer Networks, Artificial Intelligence & Soft Computing, Web Mining & Information Retrieval, Multimedia Technologies, Software Engineering and Computer Security.

The success of a Department is best judged by the performance of its students. The students find placement in prestigious organizations both at national as well as international level. Three of the students were selected for the elite civil services of Government of India. Students have fared commendably well in various national as well as international level competitions such as GATE, JRF, CAT, GRE, SAT etc. A significant number of students secure admission to higher studies at institutions of international repute.

VISION AND MISSION OF THE INSTITUTE

VISION

To become an institute of excellence in scientific and technical education and research with standards at par with national and international institutes of repute and to serve as quality human resource provider to the society and industry.

MISSION

1. To offer state-of-the-art undergraduate, postgraduate and doctoral programmes.

2. To make policies and atmosphere to attract and retain best faculty.

3. To create an ambience in which new ideas and cutting-edge research flourish through effective curriculum and infrastructure so as to produce the leaders and innovators tomorrow.

4. To produce ethically strong and morally elevated human resource to serve mankind.

5. To undertake collaborative projects and consultancy for long term interaction with the academia and industry.

6. To be among top ten engineering institutes of India.

VISION AND MISSION OF THE DEPARTMENT

VISION

To achieve the status of being one of the best departments in terms of quality of research and technical manpower in the area of computer engineering

MISSION

1. To offer state-of-the-art undergraduate, postgraduate, and doctoral programmes in computer engineering.

2. To provide one of the best working environments to motivate faculty and students to work towards vision of the department and to attract best faculty and students.

3. To develop linkages with industry, other universities/institutes/research laboratories and work in collaboration with them.

4. To use our expertise in computer engineering discipline for helping society in solving problems.

B.TECH

PROGRAMME EDUCATIONAL OBJECTIVES

AND PROGRAM OUTCOMES

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

PEO1. To produce engineering graduates who shall excel in a career utilizing their education in computer engineering

PEO2. To equip graduates with curiosity so that they may continue to enhance their knowledge

PEO3. To produce graduates who are effective in multidisciplinary and diverse professional environment

PEO4. To produce graduates who should provide leadership and demonstrate professional integrity

PROGRAM OUTCOMES (POS)

PO-1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO-2: Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PO-3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO-4: Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO-5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO-6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO-7: Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO-8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO-9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO-10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO-11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO-12: Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Title — COMPUTER PROGRAMMING LABORATORY

Course Number : COA1910/CO191
Credits : 1.5
Course Category : ESA
Pre-requisite(s) : -
Contact Hours (L-T-P) : 0-0-3
Type of Course : Practical

Course Objective

To make students of all branches of B. Tech familiar with the programming concepts and to implement the algorithmic approach of problem solving in C language to gain working knowledge of C programming.

Course Outcome

Students will be able to:

1. Understand programming concepts and C language constructs such as operators and data types, control statements, functions etc.
2. Gain algorithm development skills
3. Implement programming problems in C Language

Syllabus

Introduction to Programming Environment, experiments to be conducted in the laboratory consist of, but not limited to, the following:

- Practice of Turbo C as the development environment
- Simple introductory algorithms and programs for getting input, printing formatted output etc.
- Programs introducing elementary C concepts, like variable and names
- Programs using operators
- Programs using control structures
- Programs for repetitive tasks and iterations
- Programs on arrays and strings
- Programs introducing the use of function calls
- Programs introducing basic concept of file handling and storage classes

Books:

1. Kernighan, Brian W., and Dennis M. Ritchie. "The C programming language. "*Prentice-Hall, Englewood Cliffs, New Jersey (I978).*
2. Gottfried "Theory and Problem of Programming with C" Schaum's Outline Series, TMC (Text book)
3. E. Balagurusamy "Programming in ANSI C", McGraw Hill Education India Private Limited (2016) (Text Book)

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3				1				2			2
2	2	2	1		3							3
3	3	2	1		2			2	1	1		2

Course Title COMPUTER PROGRAMMING LAB

Course Number	:	COA1911/COA1912
Credits	:	2
Course Category	:	ESA
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Practical

Course Objective

The objective of this course to make student understand Python programming basics and paradigm. To learn and understand python looping, control statements and string manipulations.

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Understand the basic concepts of scripting and the contributions of scripting language
2. Learn the basics of programming constructs, like conditions, loops, functions, etc., and sequence types in Python like Lists, Tuples, Sets and Dictionaries.
3. Design structured programs using functions in Python.
4. Develop simple applications using Python packages.

Syllabus:

Programs covering following topics:

IDLE Python Development Environment, Variables, Identifiers, Literals, Data types, Operators, Expressions, Boolean, Sets, Lists, Tuples, Dictionaries, Casting. User input. Selection control: if-else statements. Iterative control: while loop, do-while loop, for loop, Functions, Strings, Arrays, Objects/Classes. File Handling, Modules: Numpy, Pandas, Scipy, Matplotlib.

Sample List of Experiments for the Lab

1. Write a program to demonstrate different numeric data types in Python.
2. Write a program to perform different Arithmetic Operations on numbers in Python.
3. Write a program to create, concatenate and print a string and accessing sub-string from a given string.
4. Write a program to create, append, and remove lists in python.
5. Write a program to demonstrate working with tuples in python.
6. Write a program to demonstrate working with dictionaries in python.
7. Write a python program to find largest of three numbers.
8. Write a Python program to convert temperatures to and from Celsius, Fahrenheit.
9. Write a Python program to construct the following pattern, using a nested for loop.

```
* * * * * *
* * * * *
* * * *
* * *
* *
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

10. Write a Python program that prints prime numbers less than the given number.
11. Write a python program to find factorial of a number using Recursion.
12. Write a program that accepts the lengths of three sides of a triangle as inputs. The program output should indicate whether or not the triangle is a right triangle (Recall from the Pythagorean Theorem that in a right triangle, the square of one side equals the sum of the squares of the other two sides).
13. Write a python program to define a module to find Fibonacci Numbers and import the module to another program.
14. Write a python program to define a module and import a specific function in that module to another program.
15. Write a script named copyfile.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file.
16. Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order.

17. Write a Python program to convert an integer to a roman numeral.
18. Write a Python program to implement pow(x, n)
19. Write a Python program to reverse a string word by word.
20. Develop simple applications in Python

Books/References:

1. **Charles Dierbach, *Introduction to Computer Science using Python: A Computational Problem-Solving Focus*, John- Wiley and Sons, 2012.**
2. **Peter Wentworth, Jeffrey Elkner, Allen B. Downey, and Chris Meyers, *How to Think Like a Computer Scientist: Learning with Python,* 2012, Published As Online Open Book**

***Python Tutorial from W3School,* https://www.w3schools.com/python/default.asp**

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3				1				2			2
2	2	2	1		3							3
3	3	2	1		2			2	1	1		2
4	3	2	2	1	2				2			1

Course Title — OBJECT ORIENTED PROGRAMMING

Course Number : COC2032
Credits : 4
Course Category : PC (Program Core)
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Differentiate among various programming paradigms.
2. Understand and apply Object Oriented Programming Concepts such as Classes and Objects, Inheritance, Polymorphism and Exception Handling in C++.
3. Use pointers for dynamic memory management and dynamic binding.
4. Design templates for classes and functions including use of Standard Template Library (STL).

Syllabus

UNIT I INTRODUCTION
Introduction to programming and programming paradigms; Difference between object-oriented and imperative programming style; C and C++ Programming basics, Types and Declarations, Pointers, Function prototypes, and Namespaces.

UNIT II OBJECT-ORIENTED PROGRAMMING IN C++
Introduction to Object Oriented Programming: Objects, Classes, Encapsulation, Data Hiding, Polymorphism, Inheritance, Abstract Classes, Class Members, Overloading, Overriding; Program structure, Class, Derived class, Overloading

UNIT III ADVANCED TOPICS IN C++
Abstract Data Types, Virtual function, Polymorphism, Static and dynamic binding, Run-time type information, Virtual Method table; Streams and Files and Multifile Programs

UNIT IV GENERIC PROGRAMMING AND STANDARD TEMPLATE LIBRARY (STL)
Generic Programming: Templates: Function Templates, Class Templates, Exception Handling, etc; STL: Introduction, Containers, Algorithms, Iterators, Potential Problems with the STL

References

1. Robert Lafore; “Object Oriented Programming in C++; Pearson Education, 4th Edition, 2005
2. Bjarne Stroustrup, “The C++ Programming Language”, Pearson Education, 3rd Edition, 2017
3. Bruce Eckel; “Thinking in Java”; Prentice Hall

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	1							1			1
2	3	3	2		2				1			1
3	3	2	2						1			1
4	3	2	2						1			1

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions (4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society (7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication (11) Project Management and Finance (12) Life-long Learning

Course Title — OBJECT ORIENTED PROGRAMMING

Course Number : COC2030/CO203
Credits : 4
Course Category : DC (Departmental Core)
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objective

The objective of this course is to have students learn basic concepts of object-oriented programming and its difference from procedural programming, implementation of OO concepts and techniques and writing simple application, in both C++ and JAVA.

Course Outcome

Students will be able to:

1. Select suitable programming paradigm for a given problem
2. Learn object oriented programming concepts and features of C++ and Java
3. Apply object oriented concepts in a proposed solution
4. Implement a given solution in C++ and Java

SYLLABUS:

UNIT-I Introduction to Programming

Programming paradigms: Imperative, object oriented, functional and logical programming, Introduction to Object Oriented Programming: Objects, Classes, Encapsulation, Data Hiding, Polymorphism, Inheritance, Abstract Classes, Class Members, Overloading, Overriding

UNIT-II Object Oriented Programming in C++

Program structure, Class, Derived class, Overloading, Template, Exception handling, Scope operator, Virtual function

UNIT-III Some Advanced Concepts in OOP

Abstract Data Types, Pointers, Virtual function, Polymorphism, Generic Programming: Templates: Function Templates, Class Templates, Exception Handling etc.

UNIT-III Object Oriented Programming in Java

Program Structure, Java Architecture, Statements, Classes, Inheritance, Polymorphism, Arrays, API and Packages, GUI in Java

Reference:

1. Robert Lafore; “Object Oriented Programming in C++; Pearson Education
2. Bruce Eckel; “Thinking in Java”; Prentice Hall

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3										3
2	3				1							
3	3	3	3	2	3	1						
4	3	1	3	1	3							

Course Title DATA STRUCTURE AND ALGORITHM

Course Number	:	COC2062
Credits	:	4
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Understand the basics of Data Structures and Algorithms including Asymptotic Notations.
2. Apply Linear Data Structures such as Array, Linked List, Stack, Queue in problem solving.
3. Apply Non-Linear Data Structures such as Tree and Graph in problem solving.
4. Analyze, select and use various searching and sorting algorithms.

Syllabus

UNIT I INTRODUCTION TO DATA STRUCTURES

Concept of Data Structures, Basic Terminologies related to data structures, Abstract Data Types, linear and non-linear data structure. Steps to develop an algorithm, Concept and properties of algorithms, Algorithm analysis; Complexity, Time-Space Tradeoff, Rate of growth: Big Oh notation, other asymptotic notations for complexity of algorithms.

UNIT II LINEAR DATA STRUCTURES

Arrays: representation, operations, and application. Multi-dimensional arrays. Linked Lists, operations on Linked Lists, Circular linked lists, doubly linked lists, Stacks: representation & operations, applications: polish notations etc., Recursion. Queue: representation & operations, Circular Queue, Priority Queue, Deque, applications.

UNIT III NON-LINEAR DATA STRUCTURES

Tree: Basic terminologies, Binary Tree: representation, in-order, preorder, and post-order traversal, Different types of binary trees: 2-Tree, binary search tree, balanced binary search tree, Heap tree, trie. Graph: representation, algorithms, and applications.

UNIT IV SEARCHING & SORTING ALGORITHMS

Searching: Linear search, Binary search, Hashing, Searching using binary search tree; Sorting: Bubble sort, Selection sort, Insertion sort, Merge sort, Quicksort, Heapsort, Counting Sort, Radix sort, other sorting methods and their analysis.

References

1. **Aaron M. Tenenbaum, Langsam *"Data Structure using C",* Pearson, 2008**
2. **Lipschutz, *"Data Structures"*, Tata McGraw Hill.**
3. **Goodrich M. Tamassia R., "*Data Structures and Algorithms in Java*", 3rd ed. Wiley**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1			2			1	1			1
2	3	2	1	1				1	1			1
3	3	2	1	1				1	1			1
4	3	3		1	2			1	1			1

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions
(4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society
(7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication
(11) Project Management and Finance (12) Life-long Learning

Course Title DATA STRUCTURE AND ALGORITHM

Course Number	:	COC2060/CO206
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To introduce the concept of data structures including arrays, linked lists, stacks, queues, binary trees, heaps, binary search trees, and graphs etc., and apply these data structures in problem solving. To introduce applications of various data structures and its use in a manner that adds to the efficiency of an algorithm in writing effective programs.

Course Outcome

The student will be able to

1. Learn how the choice of data structures and algorithm design methods impact the performance of programs.
2. Analyze the importance and use of Abstract Data Types (ADTs)
3. Design and implement elementary Data Structures such as arrays, trees, Stacks, Queues, and Hash Tables.
4. Identify algorithms as a pseudo-code to solve some common problems.
5. Explain best, average, and worst-cases of an algorithm using Big-O notation.

Syllabus:

UNIT I : INTRODUCTION
Concept of Data Structures, Basic Terminologies related to data structures, linear and non-linear data structure. Concept and properties of algorithms, How to develop an algorithm, Complexity, Time-Space Tradeoff, Algorithm analysis, Rate of growth: Big Oh notation, other asymptotic notations for complexity of algorithms.

UNIT II: ARRAYS
Arrays, one-dimensional arrays: traversal, selection, searching, insertion and deletion. Sorting: Bubble sort, selection sort, insertion sort, merge sort, quicksort, other sorting methods and their analysis. Multi-dimensional arrays, Representation of arrays in physical memory, Application of arrays.

UNIT III: Abstract Data Types (ADTs)
Abstract Data Types, Stacks, Applications of Stacks - prefix and postfix notations, Queue, Circular Queue, Priority Queue, Deque, Linked Lists, Operations on Linked Lists, Circular linked lists, doubly linked lists, concept of dummy nodes.

UNIT IV: Trees & Graphs
Basic terminologies, Binary Tree, representation and traversal of binary tree; in-order, preorder, and post-order traversal. Different types of binary trees: binary search tree, Heap trees and its application to sorting. Graph, representation and its applications. Other related topics.

Books:

1. Aaron M. Tenenbaum, Langsam "Data Structure using C", Pearson, 2008
2. Lipschutz, "Data structures" Tata McGraw Hill.
3. Goodrich M. Tamassia R., "Data Structures and Algorithms in Java", 3rd ed. Wiley

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	1	2								
2	2	3	1									
3	2	2	3	1	1			1	1	1		1
4	2	1		1								
5	2	3	2	2		1	1					

Course Title DIGITAL LOGIC AND SYSTEM DESIGN

Course Number	:	COC2072
Credits	:	4
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Understand Boolean logic operations, minimize and implement Boolean functions.
2. Analyze and synthesize digital combinational units.
3. Analyze and synthesize digital sequential units.
4. Design and implement simple digital system comprising control unit and data path.

Syllabus

UNIT I BOOLEAN ALGEBRA AND LOGIC GATES

Introduction, Binary numbers, Base-conversions, Octal and hexadecimal numbers, complements, binary codes, Basic Theorems and properties, Boolean functions and representation in canonical and standard forms, SOP and POS forms, other logic operations, Digital logic gates, Function minimization: Karnaugh map methods, Limitations of K-maps for larger variables, POS-simplification, NAND/NOR implementation, other 2-level implementations, Don't-care conditions, Tabular method.

UNIT II COMBINATIONAL LOGIC

Hardware aspect of arithmetic logic functions, Half-Adder, Full-Adder, Binary Adder/Subtractor, Parallel Adder, Magnitude Comparator, Demultiplexer, Multiplexer, Parity Checker/Generator, ROM, etc.

UNIT III SEQUENTIAL LOGIC

Definition and state representation of Flip-Flops, RS, D, JK-M/S, their working characteristics, State Tables, Excitation Tables and triggering. Asynchronous and Synchronous Counters-Design and Analysis, Counter Applications, Description and Operations of Shift Registers, Shift Register/Counters.

UNIT IV REGISTER TRANSFER LOGIC

Introduction to Register Transfer Logic, macro-operations and micro-operations, introduction to control unit, design and hardware specification of a simple computer, instruction fetch and execute cycle, Introduction to ALU design.

References

1. **M. Morris Mano, "*Digital Logic and Computer Design*", PHI. (Text Book)**
2. **Leach, Malvino and Saha, "*Digital Principles and Applications*", TMH. (Text Book)**
3. **Mano and Ciletti, "*Digital Design*", Pearson, 5th Ed.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1		1	1			1				
2	2	3	2	2								
3	2	3	2	2								
4	2	1	3	2	2			1			2	

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions
(4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society
(7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication
(11) Project Management and Finance (12) Life-long Learning

Course Title DIGITAL LOGIC AND SYSTEM DESIGN

Course Number	:	COC2070/CO207
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To provide introduction to fundamentals of digital logic in order to impart basic knowledge of digital circuits and their logical functions to create digital system.

Course Outcome

Students will be able to:

1. Understand Boolean logic operation, minimization of a Boolean function and its implementation.
2. Analyse and synthesize digital combinational units.
3. Analyse and synthesize digital sequential units.
4. Design and implement simple digital system comprising control unit and data path.

Syllabus

UNIT I. BOOLEAN ALGEBRA AND LOGIC GATES
Introduction, Binary numbers, Base-conversions, Octal and hexadecimal numbers, complements, binary codes, Basic Theorems and properties, Boolean functions and representation in canonical and standard forms, SOP and POS forms, other logic operations, Digital logic gates, Function minimization: Karnaugh map methods, Limitations of K-maps for larger variables, POS-simplification, NAND/NOR implementation, other 2-level implementations, Don't-care conditions, Tabular method.

UNIT II. COMBINATIONAL LOGIC
Hardware aspect of arithmetic logic functions, Half-Adder, Full-Adder, Binary Adder/Subtractor, Parallel Adder, Magnitude Comparator, Demultiplexer, Multiplexer, Parity Checker/Generator, ROM, etc.

UNIT III. SEQUENTIAL LOGIC
Definition and state representation of Flip-Flops, RS, D, JK-M/S, their working characteristics, State Tables, Excitation Tables and triggering. Asynchronous and Synchronous Counters-Design and Analysis, Counter Applications, Description and Operations of Shift Registers, Shift Register/Counters.

UNIT IV. REGISTER TRANSFER LOGIC
Introduction to Register Transfer Logic, macro-operations and micro-operations, introduction to control unit, design and hardware specification of a simple computer, instruction fetch and execute cycle, Introduction to ALU design.

Books:

*1. M. Morris Mano, "Digital Logic and Computer Design", PHI.
*2. Digital Principles and Applications by Leach, Malvino and Saha TMH
3. Digital Design by Mano and Ciletti, Pearson, 5th Ed.

*Text Book

Table: Mapping of Program Outcomes and Course Outcomes for COC207/CO207

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1	-	1	1	-	-	1	-	-	-	-
2	2	3	2	2	-	-	-	-	-	-	-	-
3	2	3	2	2	-	-	-	-	-	-	-	-
4	2	1	3	2	2	-	-	1	-	-	2	-

Course Title COMPUTER ARCHITECTURE

Course Number	:	COC2082
Credits	:	4
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	COC2072
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Understand the representation of numbers in a computer and design the digital components for arithmetic, logic and other operations.
2. Explain, design, and evaluate the components of memory, I/O, and processor.
3. Design the hardwired and microprogrammed control of a computer system.
4. Apply advanced computing concepts on computer design.

Syllabus

UNIT I INTRODUCTION TO BASIC COMPUTER ARCHITECTURE

Philosophy of digital computers, Von Neumann and Harvard architectures. Division and multiplication of signed and unsigned numbers. IEEE 754 standard of floating point numbers. Floating-point arithmetic operations. Method of register, bus and memory transfer. Micro-operations: arithmetic, logic and shift. Introduction to execution unit: shift register design, ALU design.

UNIT II MEMORY AND PROCESSOR ORGANIZATION

Introduction to main, cache, auxiliary and virtual memory. Concept of address mapping. Instruction and interrupt cycles. Timing and control. Design of accumulator logic. Central processing unit (CPU): single accumulator, general register and stack organization. Assembly language. RISC and CISC characteristics.

UNIT III HARDWIRED & MICROPROGRAMMED CONTROL LOGIC DESIGN

Hardwired control design: computer instruction execution. Design of computer registers. Design of basic computer. PLA controller. Microprogrammed control design: basic concepts of microprogrammed control design. Microprogrammed control design of an example computer. Microprogram sequencer.

UNIT IV ADVANCED CONCEPTS IN COMPUTER ARCHITECTURE

CPU performance measures and benchmarks. Introduction to pipeline and parallel processor. Parallelism in conventional computers. Classification of parallel computers. Pipelining: general consideration and speedup. Pipeline structures: arithmetic and instruction pipelines. Introduction to different types of available computers.

References

1. **M. Morris Mano, "Computer System Architecture" Third edition, Pearson Education India, Pvt. Ltd. (Text Book)**
2. **Mohamed Rafiquzzaman and Rajan Chandra, "Modern Computer Architecture", Galgotia publication Pvt. Ltd. (Text Book)**
3. **David A. Patterson and John. L. Hennessey, "Computer Organization and Design, The Hardware/Software Interface", 3rd Edition, Morgan Kaufmann Publishers. (Text Book)**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
1	3	2	3	1		1		1	1	1		2
2	3	3	3	2		1		2	2	1		3
3	3	3	3	3		2		2	2	2		3
4	3	2		3	1	2	2	1	2	2	1	1

Course Title COMPUTER ARCHITECTURE

Course Number	:	COC2080/CO208
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	COC2070/CO207
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To know the theory of components of a computer and its operation, and to learn applying the knowledge for designing a minimal computer system.

Course Outcome

Students will be able to:

1. Explain the philosophy of digital systems
2. Interpret the micro-operations
3. Explain and design arithmetic and logic unit
4. Design the control and execution unit to execute a binary instruction
5. Acquire the knowledge of advanced concepts of performance measure and speeding-up of computing

Syllabus

UNIT I INTRODUCTION TO BASIC COMPUTER ARCHITECTURE

Philosophy of digital computers, Method of register, bus and memory transfer, Micro-operations: Arithmetic, Logic and Shift, Introduction to execution of unit-register, ALU design and shift register design, division of unsigned integers, IEEE 754 standard of floating point numbers, Floating point arithmetic operations.

UNIT II MEMORY AND PROCESSOR ORGANIZATION

Introduction to main, cache, auxiliary and virtual memory, concept of address mapping, Instruction and Interrupt cycles, Timing and Control, Design of accumulator logic, Central Processing Unit (CPU): Single accumulator, General register and Stack organization, assembly language, RISC and CISC characteristics.

UNIT III HARDWIRED & MICROPROGRAMMED CONTROL LOGIC DESIGN

Hardwired control design: Computer instruction execution, Design of computer registers, Design of basic computer, PLA Controller. Microprogrammed control design: Basic concepts of Microprogrammed Control Design, Microprogrammed Control design of an example computer, Microprogram sequencer.

UNIT IV ADVANCED CONCEPTS IN COMPUTER ARCHITECTURE

CPU performance measures and Benchmarks, Introduction to pipeline and parallel processor, Parallelism in conventional computers, classification of parallel computers, Pipelining: general consideration and speedup, pipeline structures: arithmetic and instruction pipeline, Introduction to different types of available computers.

Books:

1. *M. Morris Mano "Computer System Architecture" Third edition, Pearson Education India, Pvt. Ltd, reprint 2007
2. *David A. Patterson and John. L. Hennessey "Computer Organization and Design, The Hardware/Software Interface", 3rd Edition, Morgan Kaufmann Publishers.
3. *Mohamed Rafiquzzaman and Rajan Chandra "Modern Computer Architecture", Galgotia publication Pvt. Ltd.

*Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2	1	1	-	2	-	1	-	1	-	-
2	2	3	2	2	-	-	-	-	-	1	-	-
3	3	3	3	2	2	-	1	-	-	1	-	2
4	3	3	3	2	2	-	1	-	-	-	-	2
5	3	3	2	3	1	-	1	-	-	-	-	2

Course Title SOFTWARE ENGINEERING

Course Number	:	COC2090/CO209
Credits	:	4
Course Category	:	DC
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objectives

This course covers practices which are applied during software development. These practices help in developing large size and complex software. With concepts and knowledge gained from this course, one can easily become part of industrial software production.

Course Outcomes

Upon completing the course, the student will be:

1. Familiar with various software development process model**s,** requirement engineering concepts and software design principles.
2. Able to understand software project metrics, quality concepts and estimate effort in software development.
3. Able to understand coding practices, styles and software testing approaches.
4. Able to develop software cooperatively in a team with an understanding about software risk management, configuration management and re-engineering approaches.

Syllabus

UNIT I **INTRODUCTION TO SOFTWARE ENGINEERING AND SOFTWARE PROCESS**

Introduction to Software Engineering: Program Vs Software; Characteristics of Software; Evolution of Software Engineering, Software categories, Software Development life cycle, Software Quality. **Software Development Processes**: Waterfall model, Incremental Models – Iterative Model and RAD Model, Evolutionary Models – Prototype and Spiral Model, Component Based Development; Fourth Generation Techniques, Unified Process.

UNIT-II **SOFTWARE METRICS, REQUIREMENT ENGINEERING AND SOFTWARE PLANNING**

Introduction: Software Measurement and Metrics, Software Quality Concepts**, Requirement Engineering:** Activities and approaches, Software Requirement specification; **Software sizing approaches**: Size oriented metrics, Function oriented metric, and evaluation techniques. **Software Project Planning**: Cost Estimation and Evaluation techniques, Software Team Structure.

UNIT-III **SOFTWARE DESIGN**

Introduction to Software Design: Objectives and Principles, Module level concepts Coupling and Cohesion, Design notation and specification; Architectural Design, Component Level Design, Interface Design; Structured Design Methodology, Design Heuristics, Verification, Concepts of Object Oriented Design.

UNIT-IV **CODING, TESTING AND OTHER TOPICS IN SOFTWARE ENGINEERING**

Coding: Programming practices and styles. **Testing**: Introduction to software testing, Testing Fundamentals, Test cases and test criteria. Black box testing, White box testing: Structural testing, Code Verification, Code Coverage and Cyclomatic Complexity. **Other topics in Software engineering**: Software Risk Management, Configuration Management and Software Re-Engineering.

BOOKS:

1. Roger S. Pressman, “Software Engineering: A Practitioner's Approach”, 7th International edition, Mc Graw Hill, 2009
2. Pankaj Jalote, “An Integrated Approach to Software Engineering”, Narosa publishing House, New Delhi, 1995.
3. Fairley, R.E. “Software Engineering Concepts”, McGraw Hill, 1992.

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	-	3	1	2	-	-	-	-	-	2	-	2
2	2	3	-	1	-	2	-	-	-	2	-	-
3	2	2	3	-	3	-	-	3	2	-	1	2
4	1	-	2	-	2	-	2	-	3	1	2	3

Course Title : **SOFTWARE ENGINEERING**
Course Number : COC3162
Credits : 3
Course Category : PC
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-0-0
Type of Course : Theory

Course Outcome

Upon completing the course, students will be able to:

1. Explain various software development process models based on Plan Driven approach and Agile Development methods.
2. Analyze and model software requirements, including estimation and planning.
3. Apply standard software design techniques for development of software systems with focus on quality.
4. Apply standard coding practices, version control methods, and design comprehensive test cases using various testing techniques.

Syllabus

UNIT I SOFTWARE PROCESS

Nature of Software – Defining Software, Software Application Domains, Legacy Software. Software Engineering – A layered technology, Process Framework, Umbrella Activities. Process Models – Perspective Process Models, Specialized Process Model, The Unified Process. Agile Development – Extreme Programming, Scrum.

UNIT II REQUIREMENT ENGINEERING AND SOFTWARE PLANNING

Requirement Engineering - tasks, Eliciting Requirements – Collaborative Methods and QFD, Software Requirement Specification (SRS). Requirement Analysis Modeling – Scenario-Based Modeling, Class-Based Methods, Behavioral Modeling. Planning a software project – Software Sizing, Effort Estimation, Risk Management.

UNIT III SOFTWARE DESIGNING AND QUALITY CONCEPTS

Software Designing - Concepts, Architectural Design - Architecture views, Architecture Styles, Component Level Design – Cohesion and Coupling. Software Quality: Quality Concepts, Software Quality Assurance (SQA) – Formal Technical Review.

UNIT IV CODING, SOFTWARE TESTING AND ADVANCE CONCEPTS

Coding: Programming Principles and Guidelines, Testing: Introduction to software testing, Testing Fundamentals - Test cases, and test criteria. Black box testing, White box testing. Other topics in Software engineering: Configuration Management, and Software Re-Engineering, Defect Removal Efficiency, Reliability.

References

1. ***Roger S. Pressman & Bruce R. Maxim, "Software Engineering: A Practitioner's Approach", 9th Edition, Mc Graw Hill, 2023. (Text Book)***
2. ***Ian Sommerville. 2016. Software Engineering (10th. ed.). Addison-Wesley Publishing Company, USA.***
3. ***Pankaj Jalote, "Software Engineering – A Precise Approach", Wiley India Pvt Ltd, New Delhi, 2010.***

Course Title — DESIGN AND ANALYSIS OF ALGORITHM

Course Number	:	COC2142
Credits	:	4
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	COC2062
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Explain the techniques for analysis of algorithms.
2. Understand and design algorithms using different algorithmic design techniques.
3. Analyze different algorithms for their computational complexity.
4. Distinguish between P and NP classes of problems and understand the tractability of different problems.

Syllabus

UNIT I — INTRODUCTION

Review of algorithmic analysis, Asymptotic Notations; Solving Recurrences: Substitution methods, Recursion-Tree method and Master method.

UNIT II — ALGORITHM DESIGN TECHNIQUES

Divide and Conquer Approach, Dynamic Programming, Greedy Algorithms; other algorithm design techniques.

UNIT III — GRAPH ALGORITHMS

Breadth First Search, Depth First Search; Minimum Spanning Trees, Shortest Path Algorithms: Dijkstra's Algorithm, The Bellman-Ford Algorithm, The Floyd-Warshall's Algorithm. Max Flow Algorithms: The Ford-Fulkerson Algorithm.

UNIT IV — NP COMPLETENESS

P and NP classes, Unsolvable problems, NP Completeness and Reducibility, Circuit Satisfiability, Examples and Proofs of NP-Complete problem. Introduction to approximation algorithms.

References

1. **Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, "*Introduction to Algorithms*", Third Edition, September 2009, MIT Press USA. (Text Book)**
2. **Alfred V. Aho, John E. Hopcroft, Jeffery D. Ullman, "*The Design and Analysis of Computer Algorithms*", Addison-Wesley, 2001.**
3. **John Kleinberg, Eva Tardos, "*Algorithm Design*", Addison Wesley**
4. **Ellis Horowitz, Sartaj Sahni, "*Fundamentals of Computer Algorithms*", Galgotia Publications, 2001.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	2	-	-	-	-	-	-	-	-	-
2	3	3	2	-	-	-	-	-	-	-	-	-
3	3	3	3	-	-	-	-	-	-	-	-	-
4	3	2	3	2	-	-	-	-	-	-	-	-

Course Title — DESIGN AND ANALYSIS OF ALGORITHM

Course Number	:	COC2140/CO314
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	COC2060/CO206
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

The course addresses:

1. Designing algorithms for the problems that are encountered in the area of Computer Engineering.
2. Analysis of time and space complexity of algorithms.
3. Graph algorithms and NP Problems.

Course Outcome

Students who completed the course have demonstrated the ability to:

5. Argue about the correctness of algorithms using inductive proofs and invariants.
6. Analyze running times of algorithms using asymptotic analysis.
7. Use different paradigms such as divide-and-conquer, greedy, dynamic programming, etc.
8. Decide whether a solution to the problem exists in a reasonable amount of time. If no such solution exists, one can consider algorithms that try to solve the problem approximately.
9. Develop efficient algorithms for the problems in the area of Computer Engineering and other engineering disciplines.

Syllabus

UNIT I — **INTRODUCTION**
Review of algorithmic analysis, Asymptotic Notations; Solving Recurrences: Substitution methods, Recursion-Tree method and Master method.

UNIT II — **ALGORITHM DESIGN TECHNIQUES**
Divide and Conquer Approach, Dynamic Programming, Greedy Algorithms, Greedy versus Dynamic Programming, Miscellaneous algorithmic problems

UNIT III — **GRAPH ALGORITHMS**
Breadth First Search, Depth First Search; Minimum Spanning Trees, Shortest Path Algorithms, Dijkstra's Algorithm, The Bellman-Ford Algorithm, The Floyd-Warshall's Algorithm; Maximum Flow; Flow Networks, The Ford-Fulkerson Algorithm.

UNIT IV — **NP COMPLETENESS**
P and NP classes, Unsolvable problems, NP Completeness and Reducibility, Circuit Satisfiability, Examples and Proofs of NP-Complete problem.

Books:

*1) Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, “Introduction to Algorithms”, Third Edition, September 2009, MIT Press USA.

2) Alfred V. Aho, John E. Hopcroft, Jeffery D. Ullman, "The Design and Analysis of Computer Algorithms", Addison-Wesley, 2001.

3) John Kleinberg, Eva Tardos, "Algorithm Design", Addison Wesley

4) Ellis Horowitz, Sartaj Sahni, "Fundamentals of Computer Algorithms", Galgotia Publications, 2001.

*Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	2	-	-	-	-	-	-	-	-	-
2	3	3	2	-	-	-	-	-	-	-	-	-
3	3	2	3	-	-	-	-	-	-	-	-	-
4	3	2	3	3	1	-	1	1	1	-	-	-
5	3	2	2	3	-	1	-	-	-	-	-	1

Course Title — DATA STRUCTURE AND OOP LABORATORY

Course Number	:	COC2910/CO291
Credits	:	2
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Practical

Course Objective

The objective of this course is to learn the application of data structure in problem solving and understand the importance of object oriented features and their applications.

Course Outcome

Upon successful completion of this course, the student will be able to:

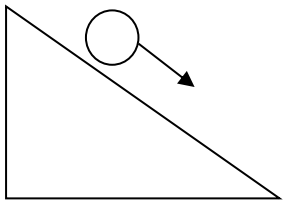
1. Design, implement and debug programs in C, C++ and Java and be able to select suitable language for a given problem
2. Design and implement data structures such as linked list, stack, tree, graphs etc. for selected solution of a given problem.
3. Apply object oriented concepts and develop object oriented solution for a given problem

Syllabus:

Programs covering following topics:

Sample List of Experiments for the Lab

1. Write a program to create and display linked list
2. Write a program to merge two linked list
3. Write a program that construct binary search tree for a given list of numbers.
4. Write a program to multiply arbitrary large numbers
5. Write a program to design a class for queue data structure
6. Write a program to overload operator for string concatenation
7. Simulate the movement of a ball if it is set free on an inclined plane (see figure).



8. Develop an object oriented solution to implement a dictionary which stores words in alphabetical order. Program takes input a word, if that word exists in the dictionary show the message ‘duplicate entry’ otherwise save that word at its correct position.
9. Construct a telephone diary using object oriented concepts. The diary contains name of person, phone number (s), Nickname, group, email address and postal address. A person can have more than one phone numbers. Perform the following operations on the telephone diary:

I. Store a new person detail
II. List the person details belonging to a given group
III. List all the phone numbers of a given person name

10. There are 16 players in a knockout tennis tournament. Players are numbered 1 through 16. In the first stage, each odd number player n plays with player number n+1. In the later stages, winner of a match plays with the winner of the next match. Simulate the tournament. Use random number generator to identify the winner in a match.

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2	2	1	3		2	2	1			1
2	2	1								1		
3	3	2	2	1	3		2	2	1			1

Course Title — DIGITAL DESIGN AND SIMULATION LAB

Course Number	:	COC2922
Credits	:	2
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	COC2072
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Lab

Course Outcome

On successful completion of the course, the student will be able to:

1. Design and implement the hardware for simple combinational and sequential circuits.
2. Use simulation tools to design complex combinational circuits.
3. Use simulation tools to design complex sequential circuits using finite state machines.
4. Design circuits to distinguish between Moore and Mealy machines.

Syllabus

Students are expected to implement various digital circuits. The outlines of the course are as follows.

Awareness of different logic gates, off-the-shelf devices, and their usage. (AND, OR, NAND, NOR, XOR, XNOR etc.).

Use of logic gates, off-the-shelf devices to implement different combinational logic circuits. (MUX, Half Adder, Full Adder, Adder/Subtractor, Registers).

Use of simulation tools to design and verify combinational and sequential circuits.

Use of memory elements to implement different Finite State Machine. (Counters, Registers, Sequence detectors).

Design, implementation and simulation of circuits using Moore and Mealy machine models with different tools.

Introduction to VHDL and Verilog.

References

1. **M. Morris Mano, "*Digital Logic and Computer Design*", PHI. (Text Book)**
2. **Leach, Malvino and Saha, "*Digital Principles and Applications*", TMH. (Text Book)**
3. **Mano and Ciletti, "*Digital Design*", Pearson, 5th Ed.**
4. **Charles H. Roth, Larry L. Kinney, "Fundamentals of Logic Design", CI-Engineering.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	1	3		1					1		2
2	2		3	2	1							1
3	3		3	1	1							1
4	3		3		1							2

Course Title DIGITAL DESIGN AND SIMULATION LAB

Course Number : COC2920
Credits : 2
Course Category : DC (Departmental Core)
Pre-requisite(s) : COC2070/CO207
Contact Hours (L-T-P) : 0-1-2
Type of Course : Lab

Course Objective

Analysis, Design and Synthesis of digital circuits on hardware and its simulation using various tools.

Course Outcome

Students who completed the course have demonstrated the ability to:

5. Understand hardware implementation of simple combinational and sequential circuits.
6. Understand simulation studies of complex digital circuits (Finite State Machine and Combinational circuits).
7. Differentiate between implementation of Moore and Mealy machines.
8. Understand the importance of timing as well as states of FSM

Syllabus

Students are expected to implement various digital circuits. The outlines of the course are as follows.

Awareness of different logic gates and its usage. (AND, OR, NAND, NOR, XOR, XNOR etc.).

Use of logic gates to implement different combinational logic circuits. (MUX, Half Adder, Full Adder, Adders/Subtractors).

Use of logic gates and memory to implement different sequential circuits. (Counters, Registers, Sequence detectors).

Design and implementation (hardware/simulation) of simple control units.

Introduction to VHDL structural Modeling Styles. Design and simulation of simple circuits using VHDL/Verilog.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	1	3		1					1		2
2	2		3	2	1							1
3	3		3	1	1							1
4	3		3		1							2

Course Title — C & C++ PROGRAMMING LAB

Course Number	:	COC2932
Credits	:	2
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	-
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Lab

Course Outcome

On successful completion of the course, the student will be able to:

1. Demonstrate proficiency in using Integrated Development Environment (IDE) to write, compile, execute and debug C/C++ programs.
2. Apply different programming concepts such as data types, control structures, functions, recursion, and file I/O operations to implement algorithms in C and C++.
3. Design and implement algorithms that involve dynamic memory allocation using pointers for efficient memory management.
4. Design and implement solution of computation problems in C++ using object-oriented programming principles such as classes, objects, inheritance, polymorphism, overloading, and template.

Syllabus

Experiments to be conducted in the laboratory consist of, but not limited to, the following:

- Exercise on learning Integrated Development Environment (IDE).
- Programs using data types including arrays and control structures
- Programs using functions and file I/O operations
- Programs involving dynamic memory allocation using pointers.
- Programs involving objects and classes
- Programs involving inheritance and polymorphism
- Programs using overloading and template

References

1. **Kernighan, Brian W., and Dennis M. Ritchie. *"The C Programming Language"*, Pearson Education.**
2. **Gottfried "Theory and Problem of Programming with C" Schaum's Outline Series, TMC**
3. **E. Balagurusamy "Programming in ANSI C", McGraw Hill Education India Private Limited (2016)**
4. **Bjarne Stroupstrup, *"The C++ Programming Language"*, Addison Wesley, Pearson Education.**
5. **Robet Lafore, "Object Oriented Programming in C++", SAMS.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1					3							3
2	3								1			
3	3								1			
4	3	3	1					2	1	1		2

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions
(4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society
(7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication
(11) Project Management and Finance (12) Life-long Learning

Course Title	**: DATA STRUCTURE LAB**
Course Number	: COC2942
Credits	: 2
Course Category	: PC
Contact Hours (L-T-P)	: 0-1-2
Type of Course	: Lab

Course Outcome

On successful completion of the course, the student will be able to:

1. Implement fundamental data structures such as arrays, linked lists, stacks, and queues in a given programming language.
2. Demonstrate proficiency in implementing advanced data structures trees, graphs and hashing using suitable programming languages.
3. Analyze computing problems and select the most appropriate data structure to solve them.
4. Design and implement custom data structures tailored to solve specific computing problems.

Syllabus

Experiments to be conducted in the laboratory consist of, but not limited to, the following:

- Programs using single dimensional and multidimensional arrays such as searching and sorting.
- Programs related to linked list such as its construction, traversal, node insertion and deletion.
- Programs on implementing and using stack and queue data structure.
- Programs on implementing and using following data structure - trees, graphs, and hashing.
- Select the most appropriate data structure for given computing problems and write program for the problem
- Design custom data structure for given computing problems and implement it.

Sample Programs

1. Search for a given number in a sorted array using (i) linear search and (ii) binary search algorithms.
2. Search for a given number in a two-dimensional array using only binary search. For each part of the program, print the number of comparisons also.
3. Sort an array of integers using standard bubble sort algorithm. Also implement a variation of bubble sort that use both “early termination” and “alternate passes in opposite direction”.
4. Sort an array of integers using standard insertion sort algorithm. Also implement a variation of insertion sort that use binary search to find the proper location to insert the next element instead of linear search.
5. Implement Merge Sort. Implementation must be non-recursive. Use tabbing with binary search in the merging process.
6. Implement hashing. Use a hash table of size 50. Use the following collision resolution methods: linear probing, quadratic probing, double hashing and separate chaining. For each of the collision resolution method, implement the following functions: a function insert (n) that adds the number n to the table, if it is not already there. If a number is already found in the hash table, it is not inserted for the second time. a function delete (n) that removes the number n from the table, if it's there. a function search(n) that returns location of number n in the table if it is present, otherwise returns ‘-1’.
7. Implement a linear list data structure using linked list representation. In a linked list representation space allocation is done to each node on demand also each node stores the pointer to the next node of linked list and its integar ‘x’ data. Implement the following functions/methods in the linked list. insert(k,x), insert a new element x just after the k th element. delete(k,x), delete the kth element and return it in x. void display(), to print the complete linked list. find(x), return the position of x in the linked list, return 0 if x is not found. bool isempty(), return true if the list is empty, false otherwise. int length(), return the number of elements in the linked list.
8. Convert a given arithmetic expression in infix notation to postfix notation. Implement the stack using singly linked list and the concept of pointers in C. Assume there are only five operators: exponent, multiplication, division, addition and subtraction.

9. Implement quicksort when pivot element is: a) The first element of the sub-array. b) Median of first, last and middle element of the sub-array. c) Randomly chosen as any element in sub-array. d) The middle element of the sub-array (if size of sub-array is even, then any of the middle two elements can be taken as pivot).
10. Implement priority queue in C using the following representations: 1. Unordered Single Array 2. Ordered linked list 3. Multiple array (Each element in the priority queue has an external numeric priority associated with it.)
11. Implement binary tree with following methods: Create(P, X) : P is location of parent of node which is to be created, X is either Left or Right. Delete(N): N is location of node which is to be deleted. If the node to be deleted is an internal node, fill the space vacated by deleted node by moving the last node (rightmost node at the last level) to the vacant space. Display () : display the entire binary tree (information and location of each node, location of its left and right children) Search(v): vis some value, search all node(s)n having value equal to v and print the location of those node(s). Also print the location of parent of those node(s). Also, implement the inorder, preorder and postorder traversal of binary tree.
12. Implementation of Binary Search Tree/Heap.
13. Implement Selection Sort & Insertion Sort in C/C++. Show the intermediate results of each pass. Also print the number of comparisons and the number of interchanges.

References

1. **Lafore, Robert. Data structures and algorithms in Java. Sams publishing.**
2. **Lipschutz, Seymour. "Data Structures with C, Schaum's Outline Series.**
3. **Tenenbaum, Aaron M. Data structures using C. Pearson Education India.**
4. **Tamassia, Roberto, and Luca Vismara. "Michael T. Goodrich." DATA STRUCTURES and APPLICATIONS.**
5. **Lafore, Robert. Sams Teach Yourself Data Structures and Algorithms in 24 Hour. Sams.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2	2	-	1	-	-	-	-	-	-	-
2	2	2	2	-	1	-	-	-	-	-	-	-
3	2	3	3	3	2	-	-	-	-	-	-	2
4	2	3	3	3	2	-	-	-	-	-	-	2

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions
(4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society
(7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication
(11) Project Management and Finance (12) Life-long Learning

Course Title DIGITAL ELECTRONICS

Course Number	:	COC3080/CO308
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	ELA1110/EL111, ELA2110/EL211
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

This course is designed to teach students the theory of digital electronics, logic and the implementation of modules required for digital systems with different cost criteria, digital integrated circuits used for building digital systems. It includes the description and the structure of various fundamental components of digital systems like logic gates, memory cells, interfacing units etc.

Course Outcome

Upon successful completion of this course, students should be able to:

1. Explain the structure and fundamental components of digital systems at transistor level with different design criteria.
2. Demonstrate a clear understanding of important concepts and logic family.
3. Design and explain different types of memory: RAM, ROM, PLA, PAL etc.
4. Describe the fundamental architecture of analog-digital interfacing units such as data converters, data acquisition system etc.

Syllabus:

UNIT I: IC LOGIC FAMILIES

Digital IC Terminology and Logic Circuit Families; MOS and CMOS Logic Circuits and Characteristics; CMOS Inverter, NAND, NOR, X-OR, X-NOR Gates; CMOS Complex Gates; CMOS Transmission Gate; CMOS Clocked S-R and D-Flip-Flops; Pseudo NMOS Logic Circuits; Pseudo NMOS Inverter and other Gates; Pass Transistor Logic (PTL) and Complementary Pass Transistor Logic (CPTL); Realization of Different Gates in PTL and CPTL; Bi-CMOS Digital Circuits; Bi-CMOS Inverter and Logic Gates; Comparison of various Logic Families.

UNIT II: MEMORY DEVICES

Memory Terminology, General Memory Operation; CPU Memory Connections; Semiconductors Memories; Types and Architecture; Memory Chip Organization; ROM -- Architecture, Addressing and Timing; MOS ROM; PROM, EPROM EEPROM (EAPROM), CD-ROM; ROM Applications; Programmable Logic Device Arrays (PAL and PLA); ROM/PLD Based Combinational Design; Semiconductor RAM -- RAM Organization; Static RAM, Dynamic RAM; DRAM Structure and Operation; Read/Write Cycles; DRAM Refreshing; Expanding Word Size and Capacity; Sequential Memories.

UNIT III: DYNAMIC LOGIC CIRCUIT AND MEMORY CELLS

Dynamic Logic Circuits; Basic Structure of Dynamic-MOS Logic Circuits; Cascading Dynamic Logic Gates; Dynamic CMOS Ratio-Less Shift Register Stage; Domino CMOS Logic; CMOS SRAM Memory Cell; One Transistor Dynamic RAM Cell; Differential Voltage Sense Amplifier and Address Decoders for SRAM and DRAM; Charge Coupled Device (CCD); Basic Operation of 3-Phase and 2-Phase CCD, CCD Memory Organization.

UNIT IV: INTERFACING WITH THE ANALOG WORLD

Principal of operation of Digital-to-Analog Converters (DACs); Basic Circuits Using Binary Weighted Resistors and R/2R Ladder; DAC specification; DAC Applications; Analog to Digital converters (ADCs); Digital Ramp ADC; Up/Down Digital Ramp ADC, (Tracking ADC), Successive Approximation ADC; Flash ADC, Dual Slope Integrated ADC, Data Acquisition, Sample and Hold Circuits: Multiplexed ADC.

Books:

*1. Ronald. J. Tocci, And Neal. S. Widmer, "Digital Systems – Principles And Applications", Eighth Edition, Pearson Education, New Delhi, 2001.

2. A.S. Sedra and K. C. Smith, "Microelectronic Circuits", Oxford University Press, Vth Edition, 2004.

3. J. Millman and Grabel, "Microelectronics", McGraw Hill, 1987.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	1	1								
2	2	1	3	1								1
3	2			1								2
4	3		2	2			1					2

Course Title	**: DIGITAL ELECTRONICS**
Course Number	: COC3082
Credits	: 3
Course Category	: PC
Pre-requisite(s)	: ELA1112, ELA2112
Contact Hours (L-T-P)	: 3-0-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Explain important concepts of logic family.
2. Analyze and design digital circuits using different Logic families at transistor level.
3. Design different types of memory.
4. Develop analog-digital interfacing circuits data converters and data acquisition.

Syllabus

UNIT I **IC LOGIC FAMILIES**

Digital IC Terminology and Logic Circuit Families; MOS and CMOS Logic Circuits and Characteristics; CMOS Inverter, NAND, NOR, X-OR, X-NOR Gates; CMOS Complex Gates; CMOS Transmission Gate; CMOS Clocked S-R and D-Flip-Flops; Pseudo NMOS Logic Circuits; Pseudo NMOS Inverter and other Gates; Pass Transistor Logic (PTL) and Complementary Pass Transistor Logic (CPTL); Realization of Different Gates in PTL and CPTL; Bi-CMOS Digital Circuits; Bi-CMOS Inverter and Logic Gates; Comparison of various Logic Families.

UNIT II **MEMORY DEVICES**

Memory Terminology, General Memory Operation; CPU Memory Connections; Semiconductors Memories; Types and Architecture; Memory Chip Organization; ROM -- Architecture, Addressing and Timing; MOS ROM; PROM, EPROM EEPROM (EAPROM), CD-ROM; ROM Applications; Programmable Logic Device Arrays (PAL and PLA); ROM/PLD Based Combinational Design; Semiconductor RAM -- RAM Organization; Static RAM, Dynamic RAM; DRAM Structure and Operation; Read/Write Cycles; DRAM Refreshing; Expanding Word Size and Capacity; Sequential Memories.

UNIT III **DYNAMIC LOGIC CIRCUIT AND MEMORY CELLS**

Dynamic Logic Circuits; Basic Structure of Dynamic-MOS Logic Circuits; Cascading Dynamic Logic Gates; Dynamic CMOS Ratio-Less Shift Register Stage; Domino CMOS Logic; CMOS SRAM Memory Cell; One Transistor Dynamic RAM Cell; Differential Voltage Sense Amplifier and Address Decoders for SRAM and DRAM; Charge Coupled Device (CCD); Basic Operation of 3-Phase and 2-Phase CCD, CCD Memory Organization.

UNIT IV **INTERFACING WITH THE ANALOG WORLD**

Principal of operation of Digital-to-Analog Converters (DACs); Basic Circuits Using Binary Weighted Resistors and R/2R Ladder; DAC specification; DAC Applications; Analog to Digital converters (ADCs); Digital Ramp ADC; Up/Down Digital Ramp ADC, (Tracking ADC), Successive Approximation ADC; Flash ADC, Dual Slope Integrated ADC, Data Acquisition, Sample and Hold Circuits: Multiplexed ADC.

References

1. ***Ronald. J. Tocci, And Neal. S. Widmer, "Digital Systems – Principles And Applications", Eighth Edition, Pearson Education, New Delhi, 2001. (Text Book)***
2. ***A.S. Sedra and K. C. Smith, "Microelectronic Circuits", Oxford University Press, Vth Edition, 2004.***
3. ***J. Millman and Grabel, "Microelectronics", McGraw Hill, 1987.***

Course Title — MICROPROCESSOR THEORY AND APPLICATION

Course Number	:	COC3090/CO309
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	COC2070/CO207, COC2080/CO208
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To develop an in-depth understanding of the operation of microprocessors, assembly language programming and interfacing.

Course Outcome

Upon successful completion of this course, students should be able to:

1. Understand the instruction set architecture of 8085 microprocessor, machine cycles, and address decoding for memory and I/O interface.
2. Understand instruction set architecture of 8086 microprocessor.
3. Write and analyze assembly language programs in 8085 and 8086.
4. Learn the concept of procedures, interrupt, and interrupt service routine.
5. Explain Input and output modes and interfacing.

Syllabus

UNIT 1 — **INTRODUCTION TO 8085 MICROPROCESSOR**
Introduction to microprocessor, microcomputer and assembly language, 8085 microprocessor architecture, 8085 based microcomputer, overview of 8085 instruction set, addressing modes. 8085 assembly language programming.

UNIT II — **BUS TIMINGS AND INTERFACING**
8085 machine cycles and bus timings. Timing Diagrams of Opcode Fetch, Memory read/write machine cycle etc. Address decoding to interface memory and I/O. Memory mapped I/O and I/O mapped I/O.

UNIT III — **x86 ARCHITECTURE AND PROGRAMMING**
Overview of 8086 microprocessor family. Internal architecture of 8086, addressing modes, instruction set of 8086, assembler directives, assembly language programming of x86 processors (analysis and coding), procedures and macros.

UNIT IV — **INTERRUPTS AND PERIPHERAL INTERFACING**
Interrupts, Interrupt service routines, interfacing peripheral devices with microprocessor. Introduction to 8253/8254 Programmable Timer/Counter, 8259 Priority Interrupt Controller, 8255 Programmable Peripheral Interface, and other peripheral devices.

BOOKS:

*1. R.L Gaonkar- “Microprocessor Architecture, Programming and Applications”, Penram International Publishing India Pvt Ltd, 5th edition.
*2. Doughlas V. Hall and SSSP Rao “Microprocessors and Interfacing- programming and hardware”, McGraw Hill Education, 3rd edition.

3. Barry B. Brey- “The Intel Microprocessors, Architecture Programming and Interfacing”,- Prentice Hall International.

* Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2							1			
2	3	2							1			
3	3	3	3	1	1	1	1	1	1			
4	3	2	2	1	1				1	2		2

Course Title	**: MICROPROCESSOR THEORY & APPLICATIONS**
Course Number	: COC3092
Credits	: 3
Course Category	: PC
Pre-requisite(s)	: COC2072, COC2082
Contact Hours (L-T-P)	: 3-0-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, students should be able to:

1. Understand the instruction set architecture of 8085 microprocessor, machine cycles, and address decoding for memory and I/O interface.
2. Understand instruction set architecture of x86 microprocessor.
3. Write and analyze assembly language programs in 8085 and x86 processors.
4. Learn the concept of procedures, interrupt, and interrupt service routine.
5. Explain Input and output modes and interfacing.

Syllabus

UNIT 1 INTRODUCTION TO 8085 MICROPROCESSOR

Introduction to microprocessor, microcomputer and assembly language, 8085 microprocessor architecture, 8085 based microcomputer, overview of 8085 instruction set, addressing modes. 8085 assembly language programming.

UNIT II BUS TIMINGS AND INTERFACING

8085 machine cycles and bus timings. Timing Diagrams of Opcode Fetch, Memory read/write machine cycle etc. Address decoding to interface memory and I/O. Memory mapped I/O and I/O mapped I/O.

UNIT III x86 ARCHITECTURE AND PROGRAMMING

Overview of x86 microprocessor family. Internal architecture of x86-64, addressing modes, instruction set, assembler directives, assembly language programming of x86 processors (analysis and coding), procedures and macros.

UNIT IV INTERRUPTS AND PERIPHERAL INTERFACING

Interrupts, Interrupt service routines, interfacing peripheral devices with microprocessor. Introduction to 8253/8254 Programmable Timer/Counter, 8259 Priority Interrupt Controller, 8255 Programmable Peripheral Interface, and other peripheral devices.

References

1. ***R.L Gaonkar- "Microprocessor Architecture, Programming and Applications", Penram International Publishing India Pvt Ltd, 5th edition. (Text Book)***
2. ***Doughlas V. Hall and SSSP Rao "Microprocessors and Interfacing- programming and hardware", McGraw Hill Education, 3rd edition. (Text Book)***

Course Title OPERATING SYSTEMS

Course Number	:	COC3100/CO310
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	COC2080/CO208
Contact Hours (L-T-P)	:	4 (3-1-0)
Type of Course	:	Theory

Course Objective

Consider and address the issues in the design of modern operating systems. The course explains the main components of modern operating system and their organization schemes. It discusses various issues and problems in the design and implementation of main components-process management, Main Memory Management, File System management, Secondary Storage Management; and methods, algorithms to solve them efficiently. Students will learn modern OS concepts through examples of Linux and Windows XP Operating Systems.

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Grasp a fundamental understanding of goals, components and evolution of operating systems
2. Learn the concepts of processes, scheduling policies, critical section problem and synchronization
3. Understand Paging, segmentation and virtual memory concepts in modern OSs
4. Understand the concepts of data input/output, storage and file management
5. Understand the main design concepts of Linux and Windows OS

Syllabus:

UNIT I **INTRODUCTION**

Evolution of Operating Systems- Simple Batch Systems, Multiprogramming system, Time-sharing systems, Parallel & Distributed systems, Real Time system; Operating System Structure: System components, Services, Interface to user & programs, concept of virtual machines, System Structure.

UNIT II **PROCESS MANAGEMENT**

Process states, control block, operations on process, Cooperating process, Concepts of Threading, IPC, Process Scheduling- Basic concept, Type of schedulers, Scheduling criteria, scheduling algorithm, Multiple processor scheduling, Real time scheduling; Process Synchronization: Critical section problem, synchronization hardware, Semaphores, classical problem of synchronization, Monitors.

UNIT III **DEADLOCK AND MEMORY MANAGEMENT**

Deadlock- Characterization, Methods of handling: Deadlock prevention, Deadlock avoidance, Deadlock detection & Recovery; Memory Management- Contiguous allocation, Swapping, Non Contiguous allocation: Paging, Segmentation, Virtual Memory- Basic Concepts, page allocation & replacement policies, Thrashing.

UNIT IV **STORAGE MANAGEMENT & CASE STUDIES**

File System- Concept, Access Methods, Directory, Protection, Implementation of File systems; Storage Structure- Overview, Disk Scheduling, Disk Management, Swap Space Management, Case Studies- Linux, Windows XP.

Books:

*1. Peterson: Silberschatz, Galvin "Operating System Concepts", Addison Wasely, Vth Addition.
2. William Stallings, "Operating Systems"
3. Milenkovic, Milan: Operating system concepts and Design, McGraw Hill.

*Text Book

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1						1	1					1
2	2	2	2	1	2			1	1			1
3	2	2	2		2							1
4	2	2	2		2							1
5			2	2	2			1	1			1

Course Title	**: OPERATING SYSTEMS**
Course Number	: COC3102
Credits	: 4
Course Category	: PC
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 3-1-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Explain the services provided by an operating system and different structures of an operating system.
2. Analyze CPU scheduling algorithms based on a scheduling criteria for selection of an appropriate scheduling algorithm.
3. Apply suitable synchronization mechanisms to solve classical problems of synchronization and avoid deadlocks during resource allocation by an operating system.
4. Evaluate memory management schemes including paging and disk scheduling schemes to choose suitable policies and algorithms.

Syllabus

UNIT I **OVERVIEW**

Introduction: What operating systems do? User's View and Systems View, Concepts of Multiprogramming and Multitasking, Virtualization, Operating System Structures: Operating System Services, User and Operating System Interface, System Services, Operating System Structures.

UNIT II **PROCESS MANAGEMENT**

Process Concepts, Process Scheduling, Operations on Processes Inter-Process Communication, Threads and Multithreading Models, CPU Scheduling, Scheduling Criteria, Scheduling Algorithms, Real Time Scheduling.

UNIT III **PROCESS SYNCHRONIZATION AND DEADLOCKS**

Process Synchronization: Critical Section Problem, Hardware Support, Semaphores, Monitors Classical Problems of Synchronization, Deadlocks: Deadlock Characterization, Methods of Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.

UNIT IV **MEMORY AND STORAGE MANAGEMENT**

Main Memory: Background, Contiguous Memory Allocation, Paging, Structure of Page Table, Swapping; Virtual Memory: Background, Demand Paging, Page Replacement, Thrashing, Operating System Examples; Mass Storage Structure: Overview, Hard Disk Drive (HDD) Scheduling.

References

1. ***Silberschatz, Galvin, and Gagne, "Operating System Concepts", Wiley India, 10th Edition. (Text Book)***
2. ***William Stallings, "Operating Systems: Internals and Design Principles", Prentice Hall, 7th Edition***
3. ***Milenkovic, Milan: Operating System Concepts and Design, McGraw Hill.***

Course Title THEORY OF COMPUTATION

Course Number	:	COC3110/CO311
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	AMS2630/AM263, COC2070/CO207
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To build a strong theoretical background for computer science, and to know practical implications of the theory

Course Outcome

Students will be able to:

1. Explain and classify mathematical models for representing finite state systems
2. Design various types of automata and write regular expressions for regular languages, and interconvert automata and regular expressions
3. Explain, analyze, and design context free grammars for context free languages
4. Design and analyze Turing machines
5. Interpret and explain the limits of computing

Syllabus

UNIT I **INTRODUCTION TO FINITE AUTOMATA & REGULAR LANGUAGES**
Strings, Alphabets and Languages, Graphs & Trees, Sequential machine, State tables & diagram, Mealy & Moore machines, State and Machine equivalence. Deterministic and Non-deterministic Finite Automata, Regular Expressions, Regular grammar, Minimization of DFA, Pumping Lemma for Regular sets, Properties of Regular Languages.

UNIT II **CONTEXT FREE GRAMMARS & LANGUAGES**
Context free Grammar, Chomsky Normal form and Greibach Normal form, Pushdown Automata, Context Free languages, Chomsky Classification of languages, Pumping Lemma for context free languages, properties of context free languages.

UNIT III **TURING MACHINES**
Turing Machines, Computing with Turing Machines, Nondeterministic Turing Machines, Church's Thesis, Universal Turing Machines.

UNIT IV **UNCOMPUTABILITY & COMPUTATIONAL COMPLEXITY**
Halting Problems, Unsolvable Problems about Turing Machines, Time bound Turing Machines, The Class P and NP Languages, NP Completeness, Some NP Complete Problems.

Books:

* 1. J.E. Hopcroft, Motwani & J.D. Ullmann, "Introduction to Automata Theory, Languages and Computation", Narosa Publications
2. H.R. Lewis & C.H. Papadimitrou, "Elements of the Theory of Computation", PHI

3. John C. Martin, "Introduction to Languages and the Theory of Computation", McGraw-Hill International
4. Michael Sipser , "Introduction to the Theory of Computation ",Thomson Learning, PWS publishing company
5. D.A. Cohen , "Introduction to Computer Theory", John Wiley
6. Zvi Kohavi, "Switching and Finite Automata Theory", Tata McGraw-Hill

(*Text Book)

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2							1			
2	3	3	3	1					1			
3	3	3	3	2	1				1			
4	3	3	3	2	1	2	1		1			
5	3	3		2	1				1			

Course Title	**: THEORY OF COMPUTATION**
Course Number	: COC3112
Credits	: 4
Course Category	: PC
Pre-requisite(s)	: AMS2632, COC2072
Contact Hours (L-T-P)	: 3-1-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Explain the concept of formal languages and design finite automata.
2. Derive regular expressions for regular languages, and convert between regular expression and finite automata.
3. Explain, analyze, and design context free grammars for context free languages
4. Design and analyze Turing machines and explain concepts of undecidability.

Syllabus

UNIT I **INTRODUCTION TO FORMAL LANGUAGES & FINITE AUTOMATA**
Strings, Alphabets and Languages, Graphs & Trees, Sequential machine, State tables & diagram, Mealy & Moore machines, State and Machine equivalence. Deterministic and Non-deterministic Finite Automata, Minimization of DFA,

UNIT II **REGULAR LANGUAGES & REGULAR EXPRESSIONS**
Regular Expressions, Regular grammar, Pumping Lemma for Regular sets, Properties of Regular Languages, Conversion of finite automata to regular languages and vice versa.

UNIT III **CONTEXT FREE GRAMMARS & LANGUAGES**
Context free Grammar, Chomsky Normal form and Greibach Normal form, Pushdown Automata, Context Free languages, Chomsky Classification of languages, Pumping Lemma for context free languages, properties of context free languages.

UNIT IV **TURING MACHINES & UNDECIDABILITY**
Turing Machines, Computing with Turing Machines, Nondeterministic Turing Machines, Church's Thesis, Universal Turing Machines. Halting Problems, Unsolvable Problems about Turing Machines, Time bound Turing Machines, The Class P and NP Languages, NP Completeness.

References

1. ***J.E. Hopcroft, Motwani & J.D. Ullmann, "Introduction to Automata Theory, Languages and Computation", Narosa Publications. (Text Book)***
2. ***Peter Linz, "Introduction to Formal Languages & Automata", 6th Edition, Jones & Bartlett.***
3. ***H.R. Lewis & C.H. Papadimitrou, "Elements of the Theory of Computation", PHI***
4. ***John C. Martin, "Introduction to Languages and the Theory of Computation", McGraw-Hill International***
5. ***Michael Sipser , "Introduction to the Theory of Computation ",Thomson Learning, PWS publishing company***
6. ***D.A. Cohen , "Introduction to Computer Theory", John Wiley***
7. ***Zvi Kohavi, "Switching and Finite Automata Theory", Tata McGraw-Hill***

Course Title — DATA BASE MANAGEMENT SYSTEM

Course Number	:	COC2122
Credits	:	4
Course Category	:	PC (Program Core)
Pre-requisite(s)	:	-
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Explain the components and models of database management systems.
2. Synthesize knowledge of relational algebra and relational calculus to solve complex query problems.
3. Design a database system using normalization forms and inspect database transactions.
4. Examine security mechanisms and query optimization techniques for database systems.

Syllabus

UNIT I — INTRODUCTION TO DATA BASE CONCEPTS

Data Base Systems (DBS) and their needs, Components of DBS, DBS architecture, Data Base Administrator and its role, Database Management Systems (DBMS) and its Components, Data Models – Classification, Physical Model- Storage structure, Indexing Techniques; Conceptual Modeling- Entity-Relationship (E/R) Model.

UNIT II — RELATIONAL MODEL

Relational Model – Definitions, Relational algebra and calculus, Integrity; Structured Query Language (SQL), Database Programming- Embedded SQL, Dynamic SQL, Procedural Language extensions to the Structured Query Language (PL/SQL).

UNIT III — NORMALIZATION, RECOVERY & CONCURRENCY

Concept of Normalization- Functional Dependencies (FD), First, Second and Third normal forms, Boyce-Codd Normal Form (BCNF), Fourth & Fifth normal forms; Recovery- Transaction, Commit and Rollback, Synchronization Points, Systems and Media recovery; Concurrency-Problems, Locking, Serializability, Deadlock.

UNIT IV — ADVANCE TOPICS IN DATABASES

Security- General considerations, Security in SQL, Views, Grant and Revoke mechanism; Query optimization, Relational Database System Design, Data Mining, Data Warehousing, Distributed Database Systems, NoSQL.

References

1. **Date, C. J. , "*Introduction To Data Base Systems*", Addison Wesley 6th ed.**
2. **Elmasri & Navathe, "*Fundamentals of Database Systems*" 5th Edition.**
3. **Abraham Silberchatz, "*Data Base System Concepts*", McGraw Hill Int.**
4. **Ullman, "*Principles of Database Systems*". Galgotia Pub.**
5. **Ivan Bayross, "*Oracle Developer 2000*", B. P. B. Publication.**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3											1
2	2	3	2		2							1
3	2	3	3		2							1
4	2	3	2	2	2							1

Course Title — DATA BASE MANAGEMENT SYSTEM

Course Number	:	COC3120/CO312
Credits	:	4
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	AMS2630/AM263, COC2060/CO206
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

This course will provide students with a general overview of databases, introducing you to database history, modern database systems, the different models used to design a database, and Structured Query Language (SQL), which is the standard language, used to access and manipulates databases.

Course Outcome

Upon successful completion of this course, the student will be able to:

5. Compare and contrast the database approach and the file system approach.
6. Explain what a database management system is as well as their components and models.
7. Identify how relational algebra / relational calculus is used to construct queries for data definition commands and data manipulation commands in SQL.
8. Understand and apply the process of normalization and design normalized relations.
9. Understand what tables, indexes, and views are as well as their importance and effect.

Syllabus

UNIT I — **INTRODUCTION TO DATA BASE CONCEPTS**
Data Base Systems and their needs, Components of DBS, DBS architecture, Data Base Administrator and his role, DBMS and its Components, Data Models – Classification, Physical Model- Storage structure, Indexing Techniques; Conceptual Modeling- E/R Model.

UNIT II — **RELATIONAL MODEL**
Relational Model – Definitions, Relational algebra and calculus, Integrity; SQL, Database Programming- Embedded SQL, Dynamic SQL, PL/SQL.

UNIT III — **NORMALIZATION, RECOVERY & CONCURRENCY**
Concept of Normalization- Functional Dependencies (FD), First, Second and Third normal forms, BCNF, Fourth & Fifth normal forms; Recovery- Transaction, Commit and Rollback, Sync. Points, Systems and Media recovery; Concurrency- problems, Locking, Serializability, Deadlock.

UNIT IV — **ADVANCE TOPICS IN DATABASES**
Security- General considerations, Security in SQL, Views, Grant and Revoke mechanism; Query optimization, Relational Database System Design, Data Mining, Data Warehousing, Parallel & Distributed Data Bases.

Books:

1. Date, C. J. : "Introd. To Data Base Systems", Addison Wesley 6th ed.
2. Elmasri & Navathe “Fundamentals of Database Systems” 5th Edition
3. Abraham Silberchatz, "Data Base System Concepts", McGraw Hill Int.
4. Ullman, "Principles of Database Systems". Galgotia Pub.
5. Ivan Bayross. "Oracle Developer 2000", B. P. B. Publication

* Text Book

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	-	-	-	2	-	-	-	-	-	-	-	-
2	-	-	-	2	3	-	-	-	-	-	-	-
3	2	3	-	3	-	-	-	-	-	-	-	-
4	3	3	3	3	3	-	-	-	-	-	-	-
5	3	3	-	3	-	-	-	-	-	-	-	-

Course Title COMPUTER NETWORKS

Course Number	:	COC3130/CO313
Credits	:	4
Course Category	:	DC
Pre-requisite(s)	:	COC2080/CO208, ELA3400/EL340
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

After taking this course the students should be able to:

1. Understand protocol layers and their service models together with delays and losses in networks.
2. Understand principles, architectures, and protocols involved in network applications.
3. Design reliable data transfer protocols and understand congestion control and flow control mechanisms.
4. Understand architecture of a router, routing algorithms, protocols at the network layer, and link layer services and protocols.

Course Outcome

Students who completed the course have demonstrated the ability to:

1. Understand protocol layers and their service models together with how to compute delays and losses in networks.
2. Understand principles, architectures, and protocols involved in network applications.
3. Design protocols for reliable data transfer and understand congestion control and flow control mechanisms.
4. Understand architecture of a router, routing algorithms, protocols at the network layer, and link layer services and protocols.

Syllabus

UNIT I INTRODUCTION

Introduction to Computer Networks, Edge and Core of the Network, Circuit Switching and Packet Switching, Protocol Layers and Their Service Models, Delays and Losses in Packet Switched Networks.

UNIT II APPLICATION LAYER

Principles of Network Applications, Application Layer Services and Protocols, the Web and HTTP, File Transfer: *FTP*; Electronic Mail in the Internet: *SMTP, IMAP, POP*; Domain Name System (DNS), Peer-to-Peer Applications.

UNIT III TRANSPORT LAYER

Transport Layer Services, Connectionless Transport: *UDP*; Principles of Reliable Data Transfer: *Go-Back-N and Selective Repeat*, Connection-Oriented Transport: *TCP*; Principles of Congestion Control, TCP Congestion Control.

UNIT IV NETWORK & LINK LAYER

Virtual Circuit and Datagram Networks, The Internet Protocol: Forwarding and Addressing, Architecture of a Router, Routing Algorithms: *Link-State, Distance Vector and Hierarchical Routing*, Routing in the Internet: *RIP, OSPF, BGP*. Link Layer Services, Multiple Access Protocols, Link-Layer Addressing: *MAC Addresses, ARP, DHCP*, Ethernet.

Books:

*1. J.F. Kurose, K.W. Ross, Computer Networking: A Top-Down Approach Featuring the Internet, 7th Edition, Pearson Education, 2017.
2. Andrew S. Tanenbaum, David J. Wetherall, Computer Networks, 5th Edition, Pearson Education, 2012.
3. K.S. Trivedi, Probability and Statistics with Reliability and Queuing and Computer Science Applications, Second Edition, John Wiley, 2002.

*Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2										1
2	2	2										1
3	2	2	3									1
4	2	2	2		2							1

Course Title	**: COMPUTER NETWORKS**
Course Number	: COC3132
Credits	: 4
Course Category	: PC
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 3-1-0
Type of Course	: Theory

Course Outcome

Students who completed the course have demonstrated the ability to:

1. Explain networking protocols and their service models together with how to compute and interpret different types of delays and losses in networks.
2. Analyze architectures and protocols involved in network applications.
3. Design reliable data transfer protocols, and evaluate congestion and flow control mechanisms.
4. Distinguish different types of routing algorithms, router architectures, link layer services and protocols.

Syllabus

UNIT I **INTRODUCTION**

Introduction to Computer Networks, The Network Edge: Access Networks, Physical Media. The Network Core: Packet Switching, Circuit Switching, A Network of Networks. Protocol Layers and Their Service Models: Layered Architecture, Encapsulation, Delays and Losses in Packet Switched Networks: Transmission, Propagation and Queuing Delays; Measuring Internet Delays: Traceroute.

UNIT II **APPLICATION LAYER**

Principles of Network Applications, Application Layer Services and Protocols, The Web and HTTP: Overview, HTTP Message Format, The User-Server Interactions: Cookies; Web Caching; File Transfer Protocol (FTP); Electronic Mail in the Internet: SMTP, Mail Access Protocols, Domain Name System (DNS): Services Provided by DNS, Working of DNS, DNS Records and Messages.

UNIT III **TRANSPORT LAYER**

Transport Layer Services, Multiplexing and Demultiplexing, Connectionless Transport: UDP, UDP Segment Structure, UDP Checksum. Principles of Reliable Data Transfer: Designing a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocols, Go Back-N, Selective Repeat. Connection-Oriented Transport: TCP, The TCP Connection, TCP Segment Structure, Telnet, Round Trip Time Estimation and Timeout, Reliable Data Transfer, Flow Control, TCP Connection Management. Principles of Congestion Control, TCP Congestion Control, Fairness Policy.

UNIT IV **NETWORK & LINK LAYER**

Overview of Network Layer, The Internet Protocol: IPv4, Addressing, IPv6, Network Address Translation (NAT), Dynamic Host Configuration Protocol (DHCP), Architecture of a Router, Routing Algorithms: Link-State and Distance Vector, Routing in the Internet: Open Shortest Path First (OSPF), Boarder Gateway Protocol (BGP). Link Layer Services, Multiple Access Protocols, Switched Local Area Networks: Link Layer Addressing: ARP/RARP and Ethernet.

References

1. ***J.F. Kurose, K.W. Ross, "Computer Networking: A Top-Down Approach", 8th Edition, Pearson Education, 2022. (Text Book)***
2. ***Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks", 6th Edition, Pearson Education, 2022.***
3. ***K.S. Trivedi, "Probability and Statistics with Reliability and Queuing and Computer Science Applications", Second Edition, John Wiley, 2002.***

Course Title — COMPUTER GRAPHICS

Course Number	:	COC3150/CO315
Credits	:	4
Course Category	:	DC
Pre-requisite(s)	:	AMS1110/AM111, AMS1120/AM112
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objectives

This objective of this course is to introduce the fundamentals of computer graphics and its underlying principles. The course aims to explain a number of algorithms that are employed to create the most basic objects in order to render a graphical scene.

Course Outcomes:

At the end of this course, the student is expected to:

1. Define computer graphics and know recent advances, display devices and processing units
2. Derive various algorithms to draw lines and other graphics primitives
3. Understand 2D and 3D viewing and geometric transformations; and classify transformation techniques
4. Learn object representation and techniques for achieving realism; design an illumination scene based on ambient, diffused and specular lighting

Syllabus

UNIT I **INTRODUCTION TO COMPUTER GRAPHICS & GRAPHICS HARDWARE**
A Survey of Computer Graphics, Video Display Devices – Raster scan and Random scan displays, Display processing units, graphics input devices, Graphics Software packages and standards.

UNIT II **POINT PLOTTING TECHNIQUES**
Coordinate systems, Line drawing algorithms: DDA, Bresenham's line drawing Algorithm, Mid-Point Circle Algorithm, Maintaining geometric properties of displayed objects, Fill area primitives, polygon fill areas, output primitives in OpenGL, point, line and character attributes, color models, antialiasing

UNIT III **GEOMETRIC & VIEWING TRANSFORMATIONS**
Basic 2-D Geometric Transformation: Translation, Rotation and Scaling, Matrix representation of transformation in homogenous co-ordinates, Composite transformations, Reflection, Shear, Geometric transformations in 3-D space. 2-D viewing, Clipping window; Normalization and viewport transformation, Clipping algorithms: Cohen-Sutherland line clipping algorithm etc; Sutherland-Hodgman Polygon clipping algorithm, Weiler and Atherton's algorithm etc., 3-D viewing Concepts: Projection Transformations, Orthogonal, Oblique parallel and Perspective projections.

UNIT IV **OBJECT REPRESENTATION AND SURFACE RENDERING**
Polyhedra, Curved surfaces, spline representations, Cubic spline interpolation methods, Bezier and B-Spline Curves, Visible surface detection algorithms: depth buffer, A- Buffer Scan-Line, depth sorting, Ray casting methods etc. Basic illumination models, Diffuse Reflection, Specular Reflection, Gouraud and Phong Surface Rendering, Ray Tracing Methods, Interactive Picture construction techniques: dragging, rubber banding etc.

Books:

1. Hearn & Baker, “Computer Graphics with OpenGL” 3rd Ed., Pearson education *
2. Foley, Dam, Feiner and Hughes “Computer Graphics: Principle and Practice”, 2nd Ed., Pearson Education
3. Edward Angel, “Interactive Computer Graphics: A Top-Down Approach Using OpenGL, 5/E, Addison-Wesley.
4. Neuman & Sproull, “Principle of Interactive Computer Graphics”, McGraw Hill Int.

***Text Book**

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	-	-	-	-	-	-	-	-	-	-
2	-	2	2	-	-	-	-	-	-	-	-	-
3	-	2	-	2	-	-	-	-	-	-	-	-
4	-	2	-	2	-	-	-	-	-	-	-	-

Course Title	**: COMPUTER GRAPHICS**
Course Number	: COC3152
Credits	: 3
Course Category	: PC
Pre-requisite(s)	: AMS1112, AMS1122
Contact Hours (L-T-P)	: 3-0-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Explain basic concepts of computer graphics, its applications, recent advances, display devices and processing units.
2. Develop and apply various algorithms to draw lines and other graphics primitives.
3. Examine 2D and 3D viewing including clipping algorithms, geometric transformations, and apply projection techniques.
4. Model the description of objects for achieving realism, and design an illumination scene based on ambient, diffused, and specular lighting.

Syllabus

UNIT I **INTRODUCTION TO COMPUTER GRAPHICS & GRAPHICS HARDWARE**

Applications of Computer Graphics, Display Devices – Raster scan and Random scan displays, Display Technologies - CRT, LCD, LED, AMOLED etc., Graphics input devices, Graphics Software packages and standards including OpenGL.

UNIT II **POINT PLOTTING TECHNIQUES**

Coordinate systems, Line drawing algorithms: DDA, Bresenham's line drawing Algorithm, Line styles – user-defined lines. Mid-Point Circle Algorithm. Interactive Picture construction techniques: dragging, rubber banding etc. Representation of Curves, Blending functions, Curve generation algorithms - Bezier and B-Spline Curves, De Casteljau's Method. Cubic spline interpolation methods, Polyhedra.

UNIT III **GEOMETRIC & VIEWING TRANSFORMATIONS**

Basic 2-D Geometric Transformation: Translation, Rotation and Scaling, Matrix representation of transformation in homogenous co-ordinates, Composite transformations, Reflection, Shear, Geometric transformations in 3-D space. 2-D viewing. Clipping window; Normalization and viewport transformation, Clipping algorithms: Cohen-Sutherland line clipping algorithm etc; Polygon clipping algorithm: Sutherland-Hodgeman, Weiler & Atherton's algorithm etc., 3-D viewing Concepts: Projection Transformations, Orthogonal, Oblique parallel and Perspective projections.

UNIT IV **OBJECT REPRESENTATION AND SURFACE RENDERING**

Visible surface detection algorithms: Depth buffer, A- Buffer Scan-Line, depth sorting, Ray casting methods etc. Basic illumination models, Diffuse Reflection, Specular Reflection, Gouraud and Phong Surface Rendering, Ray Tracing Methods.

References

1. ***Hearn & Baker, "Computer Graphics with OpenGL" 3rd Ed., Pearson education (Text Book)***
2. ***Foley, Dam, Feiner and Hughes "Computer Graphics: Principle and Practice", 2nd Ed., Pearson Education***
3. ***Edward Angel, "Interactive Computer Graphics: A Top-Down Approach Using OpenGL, 5/E, Addison-Wesley.***

Course Title — ALGORITHMS AND OPERATING SYSTEMS LAB

Course Number	:	COC3930/CO393
Credits	:	2
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Lab

Course Objectives

This course is helpful in making students understand simple algorithms and operating system concepts. This course also provides them with an opportunity to implement the algorithms and trace errors while building software.

Course Outcomes

At the end of this course, the students will be able to

1. Design simple algorithms using data structures
2. Understand the procedures defined for different operating system tasks
3. Analyze programs in order to understand their functioning
4. Apply various programming concepts to create simple software
5. Interpret various errors encountered in software development process and trace their causes.

Syllabus/ List of experiments

1. Write a program to implement the Breadth First Search (BFS) algorithm.
2. Write a program to implement the Depth First Search (DFS) algorithm.
3. Write a program to implement Prim's algorithm using Fibonacci heap for finding minimum spanning tree (MST) in a graph
4. Write a program to implement Dijkstra's single source shortest path algorithm.
5. Write a program that draws a polygon with a given set of points as corners and tests whether a given point lies within the polygon or outside the polygon or on the boundary of the polygon.
6. Write a program to find a path from a given source cell to a given destination cell in a square matrix of cell.
7. Write a program to implement the given process scheduling algorithms.
8. Write a program to implement the given page replacement algorithms.
9. Create software that implements various disk scheduling algorithms.
10. Analyze the given program/software in order to understand its functioning and bring the required changes in its functionalities.

Text Book(s)/ Reference Book(s)

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, 7th Edition, Wiley
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", Third Edition, September 2009, MIT Press USA.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	1	2		3			2	2			
2	3	2			3				2	1		
3	2	3		2	3			2	2	2		1
4	3		2		3	1		3	2	1	1	2
5	2	3		2	2				2			1

Course Title	**: ALGORITHMS & OPERATING SYSTEMS LAB**
Course Number	: COC3932
Credits	: 2
Course Category	: PC
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 0-1-2
Type of Course	: Laboratory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Analyze and implement well-known algorithms using efficient data structure.
2. Design and implement an algorithm for a given problem.
3. Implement and analyze process scheduling and resource allocation schemes for an operating system.
4. Implement and evaluate memory management and disk scheduling schemes for an operating system.

Syllabus

Experiments to be conducted in the laboratory consist of, but not limited to, the following:

- Shortest path algorithms
- Minimum spanning tree algorithms
- Algorithm for 0-1 Knapsack problem
- Maxflow algorithms
- CPU scheduling algorithms
- Deadlock avoidance algorithms
- Memory management algorithms such as page replacement policies
- Disk scheduling algorithms

Sample Programs

1. Write a program to implement Prim's algorithm using Fibonacci heap for finding minimum spanning tree (MST) in a graph
2. Write a program to implement Dijkstra's single source shortest path algorithm.
3. Write a program to compute maximum flow in a flow network using Edmond-Karp algorithm.
4. Write a program that draws a polygon with a given set of points as corners and tests whether a given point lies within the polygon or outside the polygon or on the boundary of the polygon.
5. Write a program to find a path from a given source cell to a given destination cell in a square matrix of cell.
6. Write a program to implement the given process scheduling algorithms.
7. Write a program to implement the given page replacement algorithms.
8. Create software that implements various disk scheduling algorithms.
9. Analyze the given program/software in order to understand its functioning and bring the required changes in its functionalities.
10. Write a program to implement Banker's algorithm to determine if system is in deadlock

References

1. ***Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, 7th Edition, Wiley***
2. ***Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", Third Edition, September 2009, MIT Press USA.***

Course Title MICROPROCESSOR LAB

Course Number : COC3940/CO394
Credits : 2
Course Category : DC (Departmental Core)
Pre-requisite(s) : COC3090/CO309
Contact Hours (L-T-P) : 0-1-2
Type of Course : Lab

Course Objective

To develop a practical understanding of machine- and assembly-language programming 8085 and 8086/80386 microprocessors and interfacing I/O devices with the 8085.

Course Outcome

Students will be able to:

1. Design device-independent algorithms for machine/assembly language programs
2. Implement and execute assembly language programs on 8085 microprocessor kit
3. Interface various digital devices (I/O Interfacing) with 8085 microprocessor kit
4. Implement assembly language programs for 16/32 bit Processors (8086/80386) using a suitable integrated development environment (IDE).
5. Interfacing C programs with Assembly Language

Syllabus

1. Introduction to 8085 microprocessor kit and basic programs related to registers and memory.
2. Perform arithmetic operations on 8-bit numbers.
3. Perform the arithmetic operations on n-bit string of numbers.
4. Perform searching and sorting of numbers in an array.
5. Writing programs using subroutine and stack.
6. Programs on I/O interfacing devices using 8085 kit.
7. Introduction to Visual studio for 8086/80386 assembly language program.
8. Some complex programs on 8086/80386 assembly language.
9. C Program integration with assembly language.

BOOKS:

1. R.L Gaonkar- “Microprocessor Architecture, Programming and Applications”, Wiley Eastern Ltd.
2. Douglas V. Hall & SSSP Rao- “Microprocessors & Interfacing”, (TMH Publication, 3 rd Edition, 2012)

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2	3	3	3	-	-	1	2	2	-	2
2	3	3	3	3	3	-	-	1	2	2	-	2
3	3	2	3	3	3	-	-	1	2	2	-	2
4	3	2	3	3	3	-	-	1	2	2	-	2
5	3	3	2	3	3	-	-	1	2	2	-	2

Course Title : MICROPROCESSOR & EMBEDDED SYSTEMS LAB
Course Number : COC3972
Credits : 2
Course Category : PC
Pre-requisite(s) : COC3092
Contact Hours (L-T-P) : 0-1-2
Type of Course : Laboratory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Design device-independent algorithms for machine/assembly language programs
2. Implement and execute assembly language programs on 8085 microprocessor kit
3. Interface various digital devices (I/O Interfacing) with 8085 microprocessor kit
4. Implement assembly language programs for x86 microprocessors using a suitable integrated development environment (IDE).
5. To carry out experiments related to embedded systems using VHDL/Verilog

Syllabus

Experiments to be conducted in the laboratory consist of, but not limited to, the following:

1. Introduction to 8085 microprocessor kit and hands on experiments on 8085 kits related to registers and memory.
2. Programs based on arithmetic, logic operations.
3. Writing programs using subroutine and stack.
4. Programs on I/O interfacing devices using 8085 kits.
5. Programming exercise on x86 assembly language.
6. Design and simulation of Digital functional units using different VHDL/Verilog modeling styles.
7. Design of Finite State Machine using algorithmic modeling Style.
8. Design and Simulation of Digital Circuits on FPGA using VHDL/Verilog.

References

1. ***R.L Gaonkar- "Microprocessor Architecture, Programming and Applications", Wiley Eastern Ltd.***
2. ***Douglas V. Hall & SSSP Rao- "Microprocessors & Interfacing", (TMH Publication, 3 rd Edition, 2012)***
3. ***Wayne Wolf, FPGA based digital system design, Published by Prentice-Hall.2015.***
4. ***Cem Ünsalan Yeditepe, The Ohio State University, Digital System Design with FPGA Implementation Using Verilog and VHDL, 2017, TMH Publishers***

Course Title — MINOR PROJECT LAB

Course Number	:	COC3950/CO395
Credits	:	3
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	COC2090/CO209
Contact Hours (L-T-P)	:	6 (0-0-6)
Type of Course	:	Practical

Course Objective

The objective of this lab is to make the students understand the software engineering paradigm by working on a minor project.

1. Familiarize the students with software development life cycle stages
2. Learn to apply the design methodologies
3. Learn to implement, test and deploy the project efficiently
4. Develop the understanding and use of some application development environment like .Net, JVM, Eclipse etc
5. Develop the ability to work in a team

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Apply software development practices in projects
2. Express project requirements in IEEE SRS format
3. Design using standard design methodologies
4. Implement the project using a programming language and platform
5. Test and debug the project

Syllabus:

List of some sample Minor projects completed in the Lab

- Implementation of MOST architecture for neural network design optimization
- Online examination system
- Social networking website for information sharing
- Identify trends in data using HADOOP
- Auditing software for database
- Promotional website for small scale business
- Online job search website
- Network communication on Android
- Audio/video file splitter
- Adaptive e-test
- TPO Management
- Mobile application for course management
- Dynamic news alert application on android
- Blog website for AMU
- An improved Steganography scheme
- Adaptive CPU scheduling algorithm

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3			2				3			2
2		2		2					3	3	3	2
3	2	2	3	2	2	2	2	2	3	3	3	2
4		2		2	3	2		2	3	2	2	3
5		2		1					3	3	2	2

Course Title	**: MINOR PROJECT**
Course Number	: COP3952
Credits	: 3
Course Category	: PSI
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 0-0-3
Type of Course	: Laboratory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Apply concepts from computer engineering to design and develop minor project.
2. Use modern technologies and apply modern project development practices.
3. Prepare structured project documentation including requirement specification and design document.
4. Work effectively in a team demonstrating collaboration and communication.

Sample Minor Projects

01. Online examination system
02. Social networking website for information sharing
03. Identify trends in data using HADOOP
04. Portion of an open-source project
05. Auditing software for database
06. Promotional website for small scale business
07. Online job search website
08. Network communication on Android
09. Audio/video file splitter
10. Adaptive e-test
11. TPO Management
12. Mobile application for course management
13. Dynamic news alert application on android
14. Blog website for AMU
15. An improved Steganography scheme
16. Face recognition system
17. Project related to sentiment analysis

References

1. ***Pankaj Jalote, Software Project Management in Practice, Pearson.***
2. ***Robert C. Martin, Clean Code: A Handbook of Agile Software Craftsmanship, Prentice Hall***
3. ***Randy L. Joyner, William A. Rouse & Allan A. Glatthorn, Writing the Winning Thesis or Dissertation: A Step-by-Step Guide, Corwin Press***
4. ***Qin Zhu, Mike Martin, Roland Schinzinger, Ethics in Engineering, McGraw Hill, 5th Edition.***
5. ***Relevant research papers***

Course Title ADVANCED COMPUTING LAB

Course Number	:	COC3960/CO396
Credits	:	2
Course Category	:	DC
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	0-1-2
Type of Course	:	Laboratory

Course Objectives

This course is helpful in making students learn advanced tools for software development. The student is expected to learn programming and debugging tools on windows and non-windows platforms such as Linux/Unix.

Course Outcomes

At the end of this course, the student will be able to:

CO1. Perform programming on the Linux platform.
CO2. Understand programming tools for software development.
CO3. Apply software development tools to debug software.
CO4. Develop algorithms that satisfy various constraints.
CO5. Analyze software behavior to isolate anomalies.

Syllabus/ List of experiments

1. Write a program to solve Dining Philosopher problem with 5 philosophers using semaphores. There are two pairs of Red forks and Spoons and Two pair of Green forks and Spoons. A philosopher can pick similar colored forks and spoons.
2. Write a program to find second shortest path between every pair of nodes in a Graph using threads. The number of threads will be provided by the user.
3. Write a program to find spanning tree for a given graph. Also from the removed edges find whether any other spanning tree is possible. List the other possible spanning tree.
4. Develop an Android Application for student attendance management system. (2 weeks)
5. Install a server and use a database to develop a Web application.
6. Write a program that uses Socket programming to play the tic tac toe game on the Linux platform. Both players will use separate machines.
7. Write a shell script to create two folders 'old' and 'new', also determine which file have been created earlier than a given date and also later than the given date. Place the files by maintaining their hierarchy in the newly created folders.
8. Implement a game in OpenGL (eg: Tic Tac Toe, Snakes & Ladders) etc. (2 weeks)
9. Write the unit tests for the game (using GTest in C++) and appropriate assertions in functions.

Text Book(s)/ Reference Book(s)

3. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, 7th Edition, Wiley
4. Richard Blum, Linux Command Line And Shell Scripting Bible 2nd Edition, Wiley India, 2011
5. Elliot Rusty Harold, Java Network Programming: Developing Networked Applications, 4th Edition, O'Reilly Media, 2013

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2				2							1
2					2							1
3				2	2			2	1			1
4	2	2	2	2				2	1			1
5	2	2		2								1

Course Title	**: ADVANCED COMPUTING LAB**
Course Number	: COC3962
Credits	: 2
Course Category	: PC
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 0-1-2
Type of Course	: Laboratory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Design and develop web and mobile applications using modern tools, frameworks, and technologies.
2. Design and develop concurrent solutions of problems using multithreading, synchronization and message passing mechanisms.
3. Develop applications using socket and shell programming.
4. Analyze and solve problems related to computational geometry and graphics.

Syllabus

Experiments to be conducted in the laboratory consist of, but not limited to, the following:

- Web and mobile application development
- Develop programs related to various computing techniques and algorithms such as computer networks techniques, graph algorithms etc.
- Programs involving concurrency including thread and process synchronization
- Develop an application over network using socket programming
- Write shell scripts to simplify complex sequences of commands
- Develop programs involving computational geometry and graphics

Sample Programs

1. Write a program to solve Dining Philosopher problem with 5 philosophers using semaphores. There are two pairs of Red forks and Spoons and Two pair of Green forks and Spoons. A philosopher can pick similar colored forks and spoons.
2. Write a program to find second shortest path between every pair of nodes in a Graph using threads. The number of threads will be provided by the user.
3. Develop an Android Application for student attendance management system.
4. Install a server and use a database to develop a Web application.
5. Write a program that uses Socket programming on the Linux platform.
6. Write a shell script to create two folders ‘old’ and ‘new’, also determine which file have been created earlier than a given date and also later than the given date. Place the files by maintaining their hierarchy in the newly created folders.
7. Implement a game such as Tic Tac Toe, Snakes & Ladders etc.

References

1. ***Silberschatz, Galvin, and Gagne, “Operating System Concepts”, Wiley India, 10th Edition.***
2. ***Richard Blum, Linux Command Line and Shell Scripting Bible 2nd Edition, Wiley India, 2011.***
3. ***Mendel Cooper, Advanced Bash Scripting Guide Volume-1 An in-depth exploration of the art of shell scripting.***
4. ***Elliot Rusty Harold, Java Network Programming: Developing Networked Applications, 4th Edition, O’Reilly Media, 2013.***
5. ***Donald Hearn, M. Pauline Baker, Computer Graphics C Version, Prentice Hall, 2nd Edition.***

Course Title COLLOQUIUM

Course Number	:	COC3800/CO380
Credits	:	2
Course Category	:	DC (Departmental Core)
Pre-requisite(s)	:	EZH1110/EZ111, EZH2910/EZ291
Contact Hours (L-T-P)	:	0-2-0
Type of Course	:	Seminar

Course Objective

To promote students for research and innovative work and to develop skills to present technical topics in front of high technical audience.

Course Outcome

On successful completion of the course the students will:

1. Develop their skills at giving oral research presentations and evaluating such presentations
2. Be able to write a short report upon the presentations delivered
3. Be aware of current researches in various field of computer science & engineering.
4. Be a practicing professional and sustain in academic and corporate world.

Syllabus

The colloquium is structured in manner that there is a suitable mix of presentations of the students on diversified areas of computer engineering. The colloquium will be coordinated by a faculty of the department.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1								2	2	3		1
2					3			3	2	3		1
3	3	3		2		2			2			3
4		3			3	2	3	2	2	2		3

Course Title : COLLOQUIUM
Course Number : COP3802
Credits : 2
Course Category : PSI
Pre-requisite(s) :
Contact Hours (L-T-P) : 0-2-0
Type of Course : Seminar

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Select a topic of interest in Computer Engineering including emerging areas.
2. Organize a detailed technical literature survey
3. Develop presentation and soft skills to deliver technical presentation
4. Write a technical report on the selected topic.

Syllabus

The colloquium is structured in manner that there is a suitable mix of presentations of the students on diversified areas of computer engineering.

The colloquium will be coordinated by a faculty of the department.

References

1. ***Randy L. Joyner, William A. Rouse & Allan A. Glatthorn, Writing the Winning Thesis or Dissertation: A Step-by-Step Guide, Corwin Press***
2. ***Upendra Pratap, Sakshi Ahlawat, Sushma Sharma, Research and Publication Ethics, Sultan Chand & Sons, First Edition.***
3. ***Relevant research papers and other reference materials***

Course Title INFORMATION SECURITY

Course Number	:	COC4010
Credits	:	4
Course Category	:	DC (Departmental Elective)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

To develop an understanding of information assurance as practiced in computer operating systems, distributed systems, networks and representative applications. Develop a basic understanding of cryptography, how it has evolved, and some key encryption techniques used today. Develop an understanding of security policies (such as authentication, integrity and confidentiality), as well as protocols to implement such policies in the form of message exchanges.

Course Outcome

After completion of this course, the student shall be able to:

1. Describe computer and network security fundamental concepts and principles.
2. Describe the inner-workings of popular encryption algorithms, digital signatures, certificates and anti-cracking techniques.
3. Demonstrate the ability to select among available network security technology and protocols such as IDS, IPS, firewalls, VPNs etc.
4. Describe Operating system security models E.g. UNIX, LINUX WINDOWS security and methods for web application and database security.
5. Identify ethical, professional responsibilities, risks and liabilities in computer and network environment, and best practices to write a security policy.

Syllabus

UNIT I INTRODUCTION:
Need and basic goals for computer security, security threats etc. Cryptographic building blocks: symmetric and asymmetric key cryptography. Some symmetric key algorithms.

UNIT II PUBLIC-KEY ENCRYPTION & HASH FUNCTIONS:
Public key cryptography RSA, Message Authentication, cryptographic hash functions, security of hash functions, Key management Deffie –Hellman key exchange algorithm, digital signature schemes etc., with representative applications for each.

UNIT III OPERATING SYSTEM SECURITY:
Low-level protection mechanisms, access control: models for access control, some confidentiality, integrity, and hybrid models of access control such as Bell-LaPadula, Biba, Chinese Wall etc., discretionary v/s mandatory access control. Program flaws: bugs which have security implications such as buffer overflows, race conditions etc. Malicious code: viruses, worms, Trojan horses; how they work and how to defend against them.

UNIT IV NETWORK SECURITY:

Problems in network security; kinds of attacks, PKI, key exchange protocols, example protocols such as PGP, Kerberos, IPSEC/VPN, SSL, S/MIME etc. Protocol vulnerabilities: examples of protocol vulnerabilities such as in TCP/IP, denial of service attacks etc. Tools for network security such as firewalls and intrusion detection systems.

Books

- Stallings, W. Cryptography and Network Security, Principles and Practice, 4th edition, Pearson Education , 2005.
- Network Security The Complete Reference, TATA McGraw Hill Publication., 2004
- Tanenbaum, A.S., Computer Networks, 4th edition, prentice Hall
- Stinson, D., Cryptography. Theory and practice, 2nd edition, CRC Press
- Network Security Private Communication in Public World, C. Kaufman, R. Perlman, M. Spicier, 2nd edition PHI.

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	-	1	2	1	-	-	-	-	1	1	1
2	3	-	2	2	1	-	-	-	-	1	1	1
3	3	3	2	2	3	-	-	-			2	1
4	3	-	2	2	1	-	-	-	-	1	1	1
5	3	2	2	2	2	-	-	3			2	1

Course Title — COMPILER DESIGN

Course Number : COC4060/CO406
Credits : 4
Course Category : DC
Pre-requisite(s) : AMS2630/AM263, COC3110/CO311
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objectives

This course aims to familiarize students with compiler techniques. It covers how to build compiler of programming languages. It discusses important language issues, their syntax verification and translation to machine language. But these techniques also find application in several other problems such as string parsing, translator etc.

Course Outcomes

Upon completing the course, the student will:

1. be familiar with the techniques used in lexical analysis, syntax analysis, semantic analysis, intermediate code generation, code optimization and code generation
2. be able to build compiler
3. be able to apply compiler techniques in other domain such as in building translator, parser etc.

Syllabus

UNIT-I Introduction to Compiler & Lexical Analysis

Introduction, Structure of Compiler, Elements of Programming Language, Grammar, Derivation, Syntax Tree, Parse Tree, Ambiguous Grammar, Symbol Table, Lexical Analysis – Specification and Recognition of Tokens, Lookahead Operator, Lexical Errors.

UNIT-II Syntax Analysis and Syntax Directed Translation

Syntax Analysis – Top Down Parsing, Predictive Parser, Bottom Up Parsing, SLR, Canonical and LALR Parsing, Error Handling in Top Down and Bottom Up Parsing, Syntax Directed Definition, Synthesized and Inherited Attribute, S-Attributed Definition, L-Attributed Definition.

UNIT-III Intermediate Code Generation

Type Checking, Intermediate Code Forms, Intermediate Code Generation for Arithmetic Expression, Boolean Expression, if-then-else, goto, while statements etc. Run-Time Environment

UNIT-IV Code Generation & Code Optimization

Issues in the design of Code Generator, Basic Blocks and Flow Graph, Register Allocation and Assignment. Sources of Optimization, Optimization of Basic Blocks, Global Data Flow Analysis – Reaching Definition, Available Expression, Live Variable etc. Loops in Flow Graph.

REFERENCES

1. Aho, Lam, Sethi, Ullman, “Compilers : Principles, Techniques and Tools”, Pearson Education.
2. Aho, Ullman, “ Principles of Compiler Design”, Narosa Publising House.
3. Steven S. Muchnick, “Advanced Compiler Design Implementation”, Harcourt Asia Pte Ltd.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	-	-	-	-	-	-	-	-	-	1
2	3	3	3	-	-	-	-	2	2	2	1	1
3	3	3	2	-	-	-	-	-	-	-	-	-

Course Title NETWORK AND EMBEDDED SYSTEM LAB

Course Number : COC4950/CO495
Credits : 2
Course Category : DC (Departmental Core)
Pre-requisite(s) :
Contact Hours (L-T-P) : 0-1-2
Type of Course : Lab

Course Objective

The objective of this course is:-
a) To design and simulate a network and configure various routing protocols.
b) To carry out experiments related to embedded systems using VHDL.

Course Outcome

Students who completed the course are able to:

1. Design a network and run various network protocols.
2. Carry out simulation study of various network protocols.
3. Understand the design of subsystem using HDL.
4. Design and simulate a complex digital system, using hierarchical design methodology.

Syllabus

Students are expected to perform the following sample experiments.

1. Static Routing: Consider a network with 3 routers and 2 switches. Design a network such that 3 LANs are connected to 3 routers and there is a WAN link between 2 routers. Implement static routing for the above network. Select IP addressing of your choice
2. RIP Routing: Select 2 routers and 2 switches, such that there is WAN link between the routers and the switches are connected behind the routers on a LAN. Implement RIP routing on the given network. Select IP addressing of your choice.
3. OSPF Routing: Consider a network with 4 routers. Two routers are in area 0 (core network), and 2 routers are in area 10. There are 4 LANs connected to these 4 routers. Between these 4 routers we have WAN links. Implement OSPF routing between these routers, select IP address of your choice.
4. BGP Routing: Implement the BGP protocol on a network with 3 AS in which one AS is acting as a transit AS. Choose a network of your choice.
5. Design and simulation of Digital functional units using different VHDL modeling styles.
6. Design of Finite State Machine using Algorithmic Modeling style.
7. VHDL design of Data Path using mixed design style.
8. VHDL design of control unit and simple basic microcontrollers.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	1	3	2	2	2		-	1	1			1
2	3	3	-	1	2		-	-	1			1
3	-	3	3	1	2		-	-				1
4	2	3	3	2	2		-	1	1			1

Course Title B.TECH. PROJECT PART-I

Course Number : COC4980/CO499A
Credits : 4
Course Category : DC
Pre-requisite(s) : COC2090/CO209
Contact Hours (L-T-P) : 0-0-8
Type of Course : Lab

Course Objectives

1. An opportunity to apply the acquired knowledge into practical.
2. Ability to get hands on experience and exposure.
3. Ability to acquire writing and presentation skills.
4. Ability for project planning and execution.

Course Outcomes

Students who complete this course are able to:

1. Design innovative ideas.
2. Conduct literature survey
3. Prepare good technical project reports..
4. Work and show a good team spirit.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	2	3	2	2	3	3	3	3	3	2	2
2	2	3	3	3	2	3	2	3	3	3	3	2
3	2	2	2	1	3	2	-	3	3	3	2	2
4	1	2	2	2	1	1	1	2	3	3	3	2

Course Title B.TECH. PROJECT PART-II

Course Number : COC4990/CO499B
Credits : 6
Course Category : DC
Pre-requisite(s) : COC2090/CO209
Contact Hours (L-T-P) : 0-0-12
Type of Course : Lab

Course Objectives

1. An opportunity to apply the acquired knowledge into practical.
2. Ability to get hands on experience and exposure.
3. Ability to acquire writing and presentation skills.
4. Ability for project planning and execution.

Course Outcomes

Students who complete this course are able to:

1. Implement innovative ideas.
2. Prepare and present technical project reports.
3. Write technical papers for journals and conferences.
4. Demonstrate team work and leadership.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	3	3	3	2	3	3	1	2	1
2	2	2	2	1	3	2	-	3	3	3	3	1
3	2	2	2	1	3	2	-	3	3	3	2	2
4	1	2	2	2	1	1	1	2	3	3	3	2

Course Title DISTRIBUTED AND PARALLEL SYSTEMS

Course Number	:	COE4590/CO405
Credits	:	4
Course Category	:	DE (Departmental Elective)
Pre-requisite(s)	:	COC2080/CO208, COC3100/CO311
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

This course will introduce students to fundamentals parallel and distributed programming. The course will cover the current parallel and distributed architecture, basic issues in parallel/distributed application development, parallel/distributed algorithms, data-structures and programming methodologies, and current technologies. The course will be a "hands-on" course with programming assignments and a final project.

Course Outcome

Upon successful completion of this course, the student will be able to know the following topics:

1. SISD, SIMD, MIMD and networked computers
2. Shared memory and distributed memory computers
3. Static and dynamic interconnections
4. Message passing schemes
5. Different Parallel Algorithms.

Syllabus:

UNIT I INTRODUCTION
Need for high performance computers; Limitations of Von Neumann Architecture, Pipelining, Interleaved memory Organizations, cache, Flynn's and other Classifications; Amdahls Law; Gustufson-Barsis Law; Comparison of two laws, efficiency and speed up Curves; performance measures & metrices.

UNIT II SIMD MODELS
SIMD configurations, Shared memory and mesh- Connected models, Processor organizations - mesh, shuffle, inverse shuffle, butterfly, Hypercube; Inter connection networks & performance parameters, Sorting Algorithm on SIMD models.

UNIT III PIPELINED VECTOR PROCESSOR & MULTIPROCESSORS
Vector Operations; General pipelines, concept of reservation tables; Pipelining of Vector operations; Vectorizing Compilers, case study of Vector Computers- CRAY -1, CRAY –XMP, VP 200, IRAM etc. Tightly coupled and Loosely Coupled Systems; Shared memory and message, Passing models; Properties & Characteristics of Multiprocessors, UMA and NUMA models, Task scheduling, deadlock, cache coherency/ consistency, and their solutions.

UNIT IV PARALLEL ALGORITHMS
Abstract m/c Models; RAM & PRAM Models Analysis of Parallel, Algorithms, Cost Optical algorithms, Concepts of Complexity of algorithms; NP- hard Problem etc, Searching & Sorting algorithms on PRAM Models, Practical models of parallel Computation; BSP, Log GP models, Heterogeneous processing, Parallel programming environments, concept of Grid & Cloud computing.

Books:

*1. Hwang and Briggs, "Computer Architecture and Parallel Processing", McGraw Hill International.
2. Kai Hwang , "Advanced Computer Architecture", McGraw Hill International
3. Quinn M. J., "Designing Efficient Algorithms for Parallel Computers", McGraw-Hill International
4. Saing Soo, YL. Ching, “Parallel processing and parallel Algorithms, springs Publications.

*Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	1	1	-	-	-	-	-	-	-	2
2	3	2	-	1	-	-	-	-	-	2	-	2
3	3	3	-	3	2	-	-	-	-	-	-	2
4	3	2	-	1	-	-	-	-	-	-	-	2
5	3	3	3	3	2	-	1	2	2	-	-	3

Course Title COMPUTER NETWORK DESIGN

Course Number	:	COE4510/CO451
Credits	:	4
Course Category	:	DE (Departmental Elective)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

After taking this course the students would be able to develop an in-depth understanding of the principles of network design, its applications and services.

Course Outcome

Students who completed the course have demonstrated the ability to:

1. Understand principles of network services, applications and layered architecture in data networks
2. Knowledge of computer network hardware like routers, switches, broadband routers, interfaces etc.
3. Design of routing protocols
4. Understand principles and concepts of broadband networks like DSL, Cable, Wireless etc.
.

Syllabus

UNIT I INTRODUCTION

Internetworking Concepts, Network Types, Hierarchical Design Principles, Access Layer, Distribution Layer, Core Layer, Collision/ Broadcast Domain, IP Addressing and Summarization.

UNIT II DEVICES AND CONNECTORS

Physical Media and Connectors, Ethernet Interfaces, Serial Interfaces, Routers, Switches, Hubs, Bridges, CSU/DSU, Modems, IOS, Leased Line, Configuring IP Addresses, Cable Modems, CMTS, DSL Modem, DSLAM

UNIT III ROUTING AND SWITCHING

VLAN, Trunks, Spanning Tree, Enabling Static Routes, Configuring Dynamic Routing Protocols, Default Route, LAN-to-WAN Routing, NAT, PAT, IP Traffic Management with Access Lists, Security with Firewalls, OSPF Network Design, OSPF Fundamentals, BGP Design, BGP Route Reflectors

UNIT IV NETWORK IMPLEMENTATION AND CASE STUDIES

Establishing LANs, Configuring Ethernet, Establishing Serial Point-to-Point Connections, WAN Terminology, HDLC Encapsulation, PPP Encapsulation, Internetwork Operating System CLI, Campus LAN, Internet Service Provider, Internet Data Centre, Cable Networks, DSL Networks, Wireless Networks

Books:

*1. J.F. Kurose, K.W. Ross, Computer Networking: A Top-Down Approach Featuring the Internet, 6th Edition, Pearson Education, 2012.

2. Andrew S. Tanenbaum, David J. Wetherall, Computer Networks, 5th Edition, Pearson Education, 2012.
3. K.S. Trivedi, Probability and Statistics with Reliability and Queuing and Computer Science Applications, Second Edition, John Wiley, 2002.

*Text Book

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	1	2										1
2	1	1			2							1
3	3	2	3									1
4	1	2			2							1

Course Title — MOBILE COMPUTING

Course Number	:	COE4520/CO452
Credits	:	4
Course Category	:	DE
Pre-requisite(s)	:	COC3130/CO313
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objectives

Mobile computing and wireless networks is a young and dynamic field. Ubiquitous access to information, anywhere, anyplace, and anytime, will characterize whole new kinds of information systems in the 21st century. These are being enabled by rapidly emerging wireless communications systems such as Cellular transmissions, Personal Communications Systems, Mobile IP, Wireless Local Area networks (LANs). Moreover, the next generation communication systems are expected to provide a range of services to mobile users to support voice, video, multimedia, conventional data, and Internet access in an integrated fashion.

Course Outcomes

At the end of this course students will be able to:

1. Understand the area of wireless networking and mobile computing,
2. Understand the unique network protocol challenges and opportunities presented by wireless communications and host or router mobility.
3. Apply fundamental concepts in mobile wireless systems and mobile computing, mobile IP.
4. Analyze issues associated with small handheld portable devices and design applications that can exploit mobility and location information.

Syllabus

Unit-I INTRODUCTION

Introduction to Wireless Communication Systems: Evolution of mobile radio communications, examples of wireless communication systems, PCS Architecture, Cellular Telephony, 2G, 3G and 4G Networks, AMPS, D-AMPS, GSM, CDMA

Unit-II MOBILITY MANAGEMENT

Handoff, Inter-BS Handoff, Intersystem Handoff, Roaming Management, Handoff Management, MCHO, NCHO, MAHO, Hard Handoff, Soft Handoff

Unit-III 2G/2.5G/2.75G NETWORKS

D-AMPS, Cellular Digital Packet Data, CDPD Architecture, GSM System Overview, GSM Short Message Service (SMS), GPRS Architecture, GGSN, SGSN, EDGE, Mobile Number Portability, VoIP service for Mobile Networks

Unit-IV 3G AND 4G NETWORKS

UMTS, W-CDMA, CDMA2000, Field Trials, IP for 3G, HSDPA, HSUPA, WiMAX, LTE, Advanced LTE, Circuit Switched Fallback (CSFB)

Text Book(s)/ Reference Book(s)References:

1. T.S. Rappaport, "Wireless Communication: Principles and Practice", 2nd ed, Pearson, 2002
2. Yi-Bing Lin, "Wireless & Mobile Network Architectures", John Wiley & Sons, 2001
3. Jochen Schiller, "Mobile Communication", Pearson Education
4. Dave Wisely, Philip Eardley and Louise Burness "IP for 3G - Networking Technologies for Mobile Communications", John Wiley & Sons, 2002
5. Andreas Molisch, "Wireless Communications", 2nd ed, Wiley, 2005

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	1										
2		1	1					1				2
3	3		1	1		1	2	1				1
4	1	2		1	2		1					1

Course Title VLSI DESIGN TOOLS & TECHNIQUES

Course Number : COE4420/CO442
Credits : 4
Course Category : DE (Departmental Elective)
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objective

This course is designed to teach students the theory of VLSI design flow and design automation. It includes the description and the structure of various fundamental components of digital systems like data path and controller.

Course Outcome

Upon successful completion of this course, students should be able to:

1. Understand the design flow of VLSI system, understand the architecture design and design of various blocks and sub-blocks (combinational and sequential block).
2. Explain the structure and fundamental components of VLSI physical design.
3. Understand various algorithms used for design automation.

Syllabus

UNIT I Introduction to VLSI Design; " Design Abstraction, Design Hierarchy, Design Flow, Tools foe VLSI Design, Introduction to VLSI Technology, Layout Design Rules, Stick Diagram, Transistor Parasitic, Wires and Vias".

UNIT II Logic Gate Design, Combinational Logic Network, Layout Design Methods, Standard Cell Layout Design, Combinational network Delays, Power Optimization, Sequential machines, Sequential system and Clocking Disciplines, Sequential System Design, design Validation.

UNIT III Subsystem Design, Adder, Multiplier, Graph Algorithms for Physical Design – Classes of Graph in Physical Design, Graph Problems in Physical Design, Algorithm For interval Graphs.

UNIT IV Physical Design – Partitioning- System level, board level and chip level Partitioning Floorplanning Methods, Block Placement, Routing- Global Routing and Detailed Routing.

BOOKS:

*1. Wayne Wolf, Modern VLSI Design System on Silicon, Second edition "PHI".
2. N Sherwani Algorithm for VLSI Design Automation "Springer".
3. Sabih H Gerez, Algorithm for VLSI Design Automation "J Wiley".

* Text Book

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	2	2									1
2	2	3	3									1
3	3	1	3									1
4	3		3	1								2

Course Title PRINCIPLES OF MACHINE LEARNING

Course Number	:	COE4610/CO461
Credits	:	4
Course Category	:	DE (Departmental Elective)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	3-1-0
Type of Course	:	Theory

Course Objective

The objective of this course is to introduce concepts, techniques, and algorithms in machine learning. The course will give the student the basic ideas and intuition behind modern machine learning methods as well as formal understanding of how, why, and when they work

Course Outcome At the end of this course, the student can be expected to:

1. Define what is machine learning and its application.
2. Comprehend various supervised learning algorithms and be able to select feature sets and detect anomalies in the algorithm.
3. Explain unsupervised learning algorithms and reinforcement learning techniques.
4. Be familiar with various advance topics of machine learning and their case studies.

Syllabus

UNIT 1: INTRODUCTION

Introduction to Machine Learning, Application, Recent Development, Classification of Machine learning algorithms

UNIT 2: SUPERVISED LEARNING

Supervised learning, Linear and Logistic regression, Neural Network, Back Propagation Algorithm, Generative learning algorithms, Gaussian discriminant analysis, Naive Bayes, Support vector machines, Model selection and feature selection, Ensemble methods: Bagging, boosting, Evaluating and debugging learning algorithms.

UNIT 3: UNSUPERVISED LEARNING & REINFORCEMENT LEARNING

Clustering, K-means, Mixture of Gaussians, Factor analysis, PCA (Principal components analysis), ICA (Independent components analysis), Markov Decision Process, Value iteration and policy iteration, Linear quadratic regulation (LQR), Q-learning, Value function approximation, Policy search.

UNIT 4: ADVANCE TOPICS

Introduction to Deep learning, Graphical Model, Case Studies-Application to Computer Vision, Big Data

BOOKS:

1. Machine Learning: A Probabilistic Perspective. Kevin Murphy. MIT Press, 2012.
2. Pattern Recognition and Machine Learning. Christopher Bishop. First Edition, Springer, 2006.
3. Pattern Classification. Richard Duda, Peter Hart and David Stock. Second Edition, Wiley-Interscience, 2000.
4. Machine Learning. Tom Mitchell. First Edition, McGraw-Hill, 1997

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	2	-	-	-	-	-	-	-	1
2	3	3	3	2	-	-	-	-		-	-	2
3	3	3	3	2	-	-	-	-		1	-	2
4	3	3	3	3	1	-	-	-		-	-	2

Course Title CLOUD COMPUTING

Course Number : COE4580/CO458
Credits : 4
Course Category : DE (Departmental Elective)
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objective This course will cover the study of various algorithms involved in better implementing the cloud-based systems starting through fundamentals of deployment.

Course Outcome On successful completion of this course, students will be able to:

1. Learn algorithms involved in better implementing the cloud-based systems.
2. Learn the fundamentals of deployment in cloud computing.
3. Be able to develop application in cloud environment.

Syllabus

UNIT I: Introduction: Distributed Computing and Enabling Technologies, Cloud Fundamentals: Cloud Definition, Evolution, Architecture, Applications, deployment models, and service models.

UNIT II: Virtualization: Issues with virtualization, virtualization technologies and architectures, Internals of virtual machine monitors/hypervisors, virtualization of data centers, and Issues with Multi-tenancy.

UNIT III: Interoperability and Service Monitoring: Issues with interoperability, Vendor lock-in, Interoperability approaches. SLA Management, Metering Issues, and Report generation. Resource Management and Load Balancing: Distributed Management of Virtual Infrastructures, Server consolidation, Dynamic provisioning and resource management, Resource Optimization, Resource dynamic reconfiguration, Scheduling Techniques for Advance Reservation, Capacity Management to meet

UNIT IV: Migration and Fault Tolerance: Broad Aspects of Migration into Cloud, Migration of virtual Machines and techniques. Fault Tolerance Mechanisms, Security, Advances: Grid of Clouds, Green Cloud, Mobile Cloud Computing.

Books:

1. Cloud Computing Principles and Paradigms, Rajkumar Buyya, James Broberg, Andrzej Goscinski, Wiley Publishers, 2011.
2. Cloud Computing Bible, Barrie Sosinsky, Wiley Publishers, 2010.
3. Mastering Cloud computing, Rajkumar Buyya, Christian Vacchiola, S Thamarai Selvi, McGraw Hill, 2013.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1	-	1	-	2	-	-			-	1
2	1		-	-	2	-	3			-		1
3	2	2	2	1	-	-	-	2	1	-		2

Course Title PRINCIPLES OF CYBER PHYSICAL SYSTEMS

Course Number : COE4600/CO460
Credits : 4
Course Category : DE (Departmental Elective)
Pre-requisite(s) :

Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objective This course aims to develop an understanding of the underlying principles behind integration of computation (software), networking, and physical processes to design useful engineering systems. It primarily deals with effective orchestration of software and physical processes.

Course Outcome On successful completion of this course, students will be able to:

1. Understand the significance of the integration of computation with the physical process.
2. Build high confidence systems with real-time and concurrent behaviors.
3. Analyze computational systems that interact with physical processes.
4. Design useful applications of cyber physical systems.

Syllabus

UNIT I: Foundations of Cyber Physical Systems, Modeling Dynamic Behaviours, Continuous Dynamics: Properties of Systems, Feedback Control, Discrete Dynamics: The notion of state, Finite-State Machines, Extended State Machines, Hybrid Systems, Composition of State Machine, Concurrent Models of Computation.

UNIT II: Models of sensors and actuators, Common Sensors, actuators, Embedded Processors, Types of Processor, Parallelism, Memory Technologies,

UNIT III: Input and Output, Real-Time Task Scheduling. Types of Real-Time tasks and their characteristics. Multitasking, Different Task Scheduling algorithms

UNIT IV: Analysis and verification: Invariants and temporal logic, Equivalence and refinement, reachability analysis and model checking, Quantitative analysis, Security and Privacy, applications of Cyber Physical Systems

Books:

1. E. A. Lee and S. A. Seshia, Introduction to Embedded Systems - A Cyber-Physical Systems Approach, Second Edition, MIT Press, 2017.
2. R.Mall, Real Time Systems: Theory and Practice, Pearson Education,2007.
3. Hermann Kopetz and Gunther Bauer, The Time-Triggered Architecture, Proceedings of the IEEE 91(1): 112 - 126, 2013.
4. Edward A. Lee. Cyber-Physical Systems: Design Challenges. IRORC 2008.

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	1	2	2	1	1	-	-	-	-	-
2	3	3	1	3	2	1	-	-	-	-	-	-
3	3	2	3	2	2	2	-	-	-	-	-	1
4	3	2	3	2	2	2						

Course Title — SELECTED TOPICS IN COMPUTER ENGINEERING-II

Course Number	:	COO4470/CO447N
Credits	:	4
Course Category	:	OE (Open Elective)
Pre-requisite(s)	:	
Contact Hours (L-T-P)	:	4 (3-1-0)
Type of Course	:	Theory

Course Objective

The objective of this course is to study the fundamental concepts of Computer networking and Database Management Systems. in this course, students will learn about protocols governing various layers of OSI and TCP/IP models like flow control, techniques for routing information in packets and data organization in relational databases and maintaining its a*tomicity, consistency, isolation, durability*.

Course Outcome

Students will be able to:

1. Define computer network concepts, OSI and TCP/IP Model.
2. Explain data link layer emphasizing on multiple access of media and collision detection.
3. Explain network layer emphasizing on routing algorithm and transport layer.
4. Describe architecture and fundamental elements of Relational Database management system.
5. Learn Normalization, Recovery and Concurrency techniques

SYLLABUS:

UNIT I **COMPUTER NETWORKS**

Protocol and Protocol Structure, Layered Architecture, ISO-OSI Reference Model, TCP/IP Reference Model, Physical and Data Layer Concepts: Transmission Media, MAC Layer, Flow Control , CSMA/CD

UNIT II **TCP/IP PROTOCOLS & ROUTING ALGORITHMS**

Networks Layer : IP, Family & Protocol, Routing Algorithms Transport layer : QOS, Connection Management; Flow Control, Buffering and Crash Recovery Introduction to Telnet, FTP, SMTP, SNMP, PPP, SLIP

UNIT III **DATA BASE MANAGEMENT SYSTEM**

Data Base Systems and their needs, Components of DBS, DBS architecture, Data Base Administrator and his role, DBMS and its Components, Relational Model: Domains, Attributes & keys, Extension, Intension, Integrity Rules and Constraints, SQL

UNIT IV **NORMALIZATION, RECOVERY AND CONCURRENCY**

Normalization & Functional Dependencies, Introduction to Recovery, Transaction, Commit, Rollback, Sync, Concurrency control problems

Books:

*1. A.S. Tanenbaum , “Computer Networks” 4th edition,
*2. Abraham Silberchatz, “ Data Base System Concepts”, McGraw Hill
3. Forouzan, “Data Communication and Networking”, Tata McGraw Hill
4. Willian Stalbings , “Data Communication and Networks”,
5. Date, C.J.: “Introduction to Data base System”, Addison Wesley
6. Elmasri & Navathe “Fundamentals of Database Systems” 5th Edition

*Text Books

Course Outcomes	**Program Outcomes (POs)**											
	1	2	3	4	5	6	7	8	9	10	11	12
1	-	2	-	1		-	-	-		-	-	1
2	2	2	-	-		-	-	-	-	1	-	-
3	2	2	-	-		-	-	-	-	1	-	-
4	2	-	3	2	-	-	-	-		1	-	1
5	-	1	2	3		-	-				-	1

Course Title	**: SELECTED TOPICS IN COMPUTER ENGINEERING-II**
Course Number	: COO4470
Credits	: 4
Course Category	: OE
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 3-1-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Define the concepts of computer networks, including the OSI and TCP/IP models and explain the characteristics, functions, and principles of the physical layer and data link layer.
2. Describe the characteristics, functions, and principles of the networking layer and transport layer.
3. Understand the fundamentals of a Relational Database Management System (RDBMS), and use SQL queries.
4. Apply the concepts of Normalization, Recovery and Concurrency techniques.

Syllabus

UNIT I **PHYSICAL AND DATA LINK LAYER**
Introduction to Computer Networks, Protocol and Protocol Structure, Layered Architecture, ISO-OSI Reference Model, TCP/IP Reference Model, Physical and Data Link Layer Concepts: Transmission Media, MAC Layer, Flow Control, CSMA/CD.

UNIT II **NETWORK AND TRANSPORT LAYER**
Network Layer: IP Family & Protocol, Routing Algorithms, Transport Layer Services, Connectionless Transport: UDP; Principles of Reliable Data Transfer: Go- Back-N and Selective Repeat, Connection-Oriented Transport: TCP; Principles of Congestion Control.

UNIT III **INTRODUCTION TO DATABASE MANAGEMENT SYSTEM**
Database Systems and their advantages, Components of DBS, DBS architecture, Database administrator and his role, DBMS and its components, Relational Model: Domains, Attributes and Keys, Integrity Rules and Constraints, SQL.

UNIT IV **NORMALIZATION, RECOVERY AND CONCURRENCY**
Normalization & Functional Dependencies, First, Second and Third normal forms, BCNF. Introduction to Recovery, Transaction, Commit, Rollback, Concurrency control problems

References

1. ***A.S. Tanenbaum, "Computer Networks" 6th edition. (Text Book)***
2. ***Abraham Silberchatz, "DataBase System Concepts", McGraw Hill. (Text Book)***
3. ***Forouzan,"Data Communication and Networking", TataMcGraw Hill.***
4. ***Date, C.J.: "An Introduction to Database System", 7th edition.***
5. ***Elmasri & Navathe, "Fundamentals of Database Systems" 7th Edition***

Course Title	**: SELECTED TOPICS IN COMPUTER ENGINEERING-II**
Course Number	: COO4472
Credits	: 3
Course Category	: OE
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 3-0-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Define the concepts of computer networks, including the OSI and TCP/IP models and explain the characteristics, functions, and principles of the physical layer and data link layer.
2. Describe the characteristics, functions, and principles of the networking layer and transport layer.
3. Understand the fundamentals of a Relational Database Management System (RDBMS), and use SQL queries.
4. Apply the concepts of Normalization, Recovery and Concurrency techniques.

Syllabus

UNIT I **PHYSICAL AND DATA LINK LAYER**
Introduction to Computer Networks, Protocol and Protocol Structure, Layered Architecture, ISO-OSI Reference Model, TCP/IP Reference Model, Physical and Data Link Layer Concepts: Transmission Media, MAC Layer, Flow Control, CSMA/CD.

UNIT II **NETWORK AND TRANSPORT LAYER**
Network Layer: IP Family & Protocol, Routing Algorithms, Transport Layer Services, Connectionless Transport: UDP; Principles of Reliable Data Transfer: Go- Back-N and Selective Repeat, Connection-Oriented Transport: TCP; Principles of Congestion Control.

UNIT III **INTRODUCTION TO DATABASE MANAGEMENT SYSTEM**
Database Systems and their advantages, Components of DBS, DBS architecture, Database administrator and his role, DBMS and its components, Relational Model: Domains, Attributes and Keys, Integrity Rules and Constraints, SQL.

UNIT IV **NORMALIZATION, RECOVERY AND CONCURRENCY**
Normalization & Functional Dependencies, First, Second and Third normal forms, BCNF. Introduction to Recovery, Transaction, Commit, Rollback, Concurrency control problems

References

1. ***A.S. Tanenbaum, "Computer Networks" 6th edition. (Text Book)***
2. ***Abraham Silberchatz, "DataBase System Concepts", McGraw Hill. (Text Book)***
3. ***Forouzan,"Data Communication and Networking", TataMcGraw Hill.***
4. ***Date, C.J.: "An Introduction to Database System", 7th edition.***
5. ***Elmasri & Navathe, "Fundamentals of Database Systems" 7th Edition***

Course Title : DATA STRUCTURES AND PROGRAMMING

Course Number	:	COA2602
Credits	:	3
Course Category	:	ESA
Pre-requisite(s)	:	---
Contact Hours (L-T-P)	:	2-1-0
Type of Course	:	Theory

Course Outcome

On successful completion of the course, the student will be able to:

1. Understand the basics of data structures and algorithms including asymptotic notations.
2. Apply linear data structures such as array, linked list, stack, queue in problem solving.
3. Apply non-Linear data structures such as tree and graph in problem solving.
4. Analyze, select and use various searching and sorting algorithms.

Syllabus

UNIT I INTRODUCTION TO DATA STRUCTURES

Review of structures and Pointers in C; Data Structures: Concept, Organization, Classification, Operation; Introduction to Abstract Data Types, Introduction to linear and non-linear data structure, Algorithms: Introduction, Definition, Time and Space Complexity, Notations (Big-O, omega, Theta)

UNIT II LINEAR DATA STRUCTURES

Arrays: representation, operations, and application, Multi-dimensional arrays.; Linked Lists: Singly Linked List, Circularly Linked, Doubly Linked, Traversal, Insertion and Deletion; Stacks: Array Representation, Linked Representation, Operations on Stacks, Polish Notations, Applications; Recursion; Queues: Array Representation, Linked Representation, Operations on Queues, Types: Circular Queue, Priority Queue, Deque, Applications.

UNIT III NON-LINEAR DATA STRUCTURES

Tree: Basic terminologies, Binary Tree: representation, in-order, preorder, and post-order traversal, Types of Binary trees; Operations on Binary Search Trees; Heaps, Operations on Heaps; Graph: Terminology, Representations, Traversal Algorithms, Applications.

UNIT IV SEARCHING & SORTING ALGORITHMS

Searching: Linear search, Binary search, Searching using Binary Search Tree, Hashing; Sorting: Bubble sort, Selection sort, Insertion sort, Merge sort, Quicksort, Heapsort, Counting Sort, Radix sort, other sorting methods and their analysis.

References

1. **Lipschutz, *"Data Structures"*, Tata McGraw Hill, Revised First edition**
2. **Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein *, "Data Structure using C",* First edition, Pearson, 2019**
3. **Goodrich M. Tamassia R., "*Data Structures and Algorithms in Java*", 7th edition. Wiley**

CO-PO Mapping

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1			2			1	1			1
2	3	2	1	1				1	1			1
3	3	2	1	1				1	1			1
4	3	3		1	2			1	1			1

(1) Engineering Knowledge (2) Problem Analysis (3) Design/Development of Solutions
(4) Conduct Investigations of Complex Problems (5) Modern Tool Usage (6) Engineer and Society
(7) Environment and Sustainability (8) Ethics (9) Individual and Team Work (10) Communication
(11) Project Management and Finance (12) Life-long Learning

Course Title : **DATA STRUCTURES AND PROGRAMMING**

Course Number : COA3600
Credits : 3
Contact Hours (L-T-P) : 2-1-0
Type of Course : Theory

Course Outcome

1. The student will be familiar with commonly used Data Structures and Algorithms.
2. The student will be able to design and develop Data Structures for a given problem.
3. The student will be able to write programs in C language to solve common problems using concept of Data Structures such as arrays, linked list, tree etc. and Algorithms.

UNIT I **Introduction to Data Structures and Algorithms**
Review of structures and Pointers in C; Data Structures: Concept, Organization, Classification, Operation; Algorithms: Introduction, Definition, Time and Space Complexity, Notations (Big-O, omega, Theta)

UNIT II **Arrays, Search and Sorting Algorithms**
Arrays: Definition, Accessing Elements, Search, Deletion, Merge; representation of Arrays, and pointers in physical Memory; Binary Search Algorithms, Sorting Algorithms: bubble Sort, Insertion Sort, Selection Sort, Merge Sort, Quick Sort.

UNIT III **Linked Lists, Stacks and Queues**
Link Lists: Singly Linked List, Circularly Linked, Doubly Linked; Traversal, insertion and Deletion; Stacks: Array Representation, Linked Representation, Operations on Stacks, Multiple Stacks, Applications;
Queues: Array Representation, Linked Representation, Operations on Queues, Types, Applications.

UNIT IV **Trees and Graphs**
Trees: Types (General, Forests, Binary, Binary Search), Operations on Binary Search Trees; Heaps: Binary, Binomial, Fibonacci; Graphs: Terminology, Representations, Traversal Algorithms, Shortest Path Algorithms, Searching; Hashing: Hash Tables, Hash Functions

Books:

*1. Lipschutz, "Data Structures with C" Tata McGraw Hill.
2. Tanenbaum, "Data Structures using C", Pearson.
3. Goodrich M. Tamassia R., "Data Structures and Algorithms in Java", 3rd ed. Wiley

* Text Book

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	2	1	-	-	-	-	-	-	-	2	-	-
2	3	3	3	2	1	-	-	2	2	2	2	1
3	3	2	3	2	1	-	-	2	1	2	2	2

Course Title	**: SELECTED TOPICS IN COMPUTER ENGINEERING-I**
Course Number	: COO4460
Credits	: 4
Course Category	: OE
Pre-requisite(s)	:
Contact Hours (L-T-P)	: 3-1-0
Type of Course	: Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Understand and apply data structures such as Arrays and Linked Lists in solving problems such as searching, sorting etc.
2. Explain the concept of Stacks, Queues and Trees and apply them in problem-solving
3. Understand the concepts of operating systems such as process and memory management techniques.
4. Apply process scheduling algorithms, page replacement and disk scheduling algorithms etc.

Syllabus

UNIT I **INTRODUCTION TO DATA STRUCTURES, ARRAYS AND LINKED LISTS**
Introduction to Data Structures, Abstract Data Types, Algorithms: Introduction, Time and Space Complexity, Big-O Notation; Representation, operations and applications of: Array, Linked List- Singly Linked List, Circular Linked List; Searching: Linear search, Binary search. Sorting techniques.

UNIT II **STACKS, QUEUES AND TREES**
Representation, operations and applications of Stack and Queue; Tree: Basic terminologies, Binary Tree: representation, in-order, preorder, and post-order traversal, Types; Operations on Binary Search Trees; Heaps, Operations on Heaps

UNIT III **INTRODUCTION TO OPERATING SYSTEM AND PROCESS CONCEPT**
Evolution of Operating Systems; Operating System Structure: System components, Services, Interface to user & programs; Process Concept, Process States, Process Control Block, Process Scheduling- Basic concept, Type of schedulers, Scheduling criteria, CPU scheduling algorithms

UNIT IV **MEMORY MANAGEMENT AND DISK SCHEDULING**
Contiguous memory allocation, Swapping, Non-Contiguous allocation: Paging, Segmentation, Virtual Memory- Basic Concepts, page allocation & replacement policies, Thrashing, Disk Structure, Disk Scheduling.

References

1. **Aho, Hopcroft, Ullman, "Data Structures and Algorithms", Pearson Education. (Text Book)**
2. **Lipschutz, "Data structures" Tata McGraw Hill. (Text Book)**
3. **Silberschatz, Galvin "Operating System Concepts", 10th ed, Addisson Wesley, 2018. (Text Book)**
4. **A.K. Sharma, "Data Structure using C", Pearson, 2011**
5. **Goodrich M. Tamassia R., "Data Structures and Algorithms in Java", 6th ed. Wiley**

Course Title : **SELECTED TOPICS IN COMPUTER ENGINEERING-I**
Course Number : COO4462
Credits : 3
Course Category : OE
Pre-requisite(s) :
Contact Hours (L-T-P) : 3-0-0
Type of Course : Theory

Course Outcome

Upon successful completion of this course, the student will be able to:

1. Understand and apply data structures such as Arrays and Linked Lists in solving problems such as searching, sorting etc.
2. Explain the concept of Stacks, Queues and Trees and apply them in problem-solving
3. Understand the concepts of operating systems such as process and memory management techniques.
4. Apply process scheduling algorithms, page replacement and disk scheduling algorithms etc.

Syllabus

UNIT I **INTRODUCTION TO DATA STRUCTURES, ARRAYS AND LINKED LISTS**
Introduction to Data Structures, Abstract Data Types, Algorithms: Introduction, Time and Space Complexity, Big-O Notation; Representation, operations and applications of: Array, Linked List- Singly Linked List, Circular Linked List; Searching: Linear search, Binary search. Sorting techniques.

UNIT II **STACKS, QUEUES AND TREES**
Representation, operations and applications of Stack and Queue; Tree: Basic terminologies, Binary Tree: representation, in-order, preorder, and post-order traversal, Types; Operations on Binary Search Trees; Heaps, Operations on Heaps

UNIT III **INTRODUCTION TO OPERATING SYSTEM AND PROCESS CONCEPT**
Evolution of Operating Systems; Operating System Structure: System components, Services, Interface to user & programs; Process Concept, Process States, Process Control Block, Process Scheduling- Basic concept, Type of schedulers, Scheduling criteria, CPU scheduling algorithms

UNIT IV **MEMORY MANAGEMENT AND DISK SCHEDULING**
Contiguous memory allocation, Swapping, Non-Contiguous allocation: Paging, Segmentation, Virtual Memory- Basic Concepts, page allocation & replacement policies, Thrashing, Disk Structure, Disk Scheduling.

References

1. ***Aho, Hopcroft, Ullman, "Data Structures and Algorithms", Pearson Education. (Text Book)***
2. ***Lipschutz, "Data structures" Tata McGraw Hill. (Text Book)***
3. ***Silberschatz, Galvin "Operating System Concepts", 10th ed, Addisson Wesley, 2018. (Text Book)***
4. ***A.K. Sharma, "Data Structure using C", Pearson, 2011***
5. ***Goodrich M. Tamassia R., "Data Structures and Algorithms in Java", 6th ed. Wiley***

Course Title : **NATURAL LANGUAGE PROCESSING**
Course Number : COE4620
Credits : 4
Course Category : DE (Departmental Elective)
Pre-requisite(s) : -
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Objective

This course provides an introduction to the field of computational linguistics, aka natural language processing (NLP). The course will cover linguistic (knowledge-based) and statistical approaches to language processing in the three major subfields of NLP: syntax (language structures), semantics (language meaning), and pragmatics/discourse (the interpretation of language in context).

Course Outcome

Students will be able to:

1. Comprehend the concept of natural language processing, its challenges and applications.
2. Acquire the knowledge of syntax and semantics related to natural languages.
3. Design and analyze various NLP algorithms.
4. Sketch and evaluate machine learning techniques used in NLP applications.

SYLLABUS:

UNIT-I Introduction and Word Level Syntactic Analysis

Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM - Regular Expressions, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance, Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff. Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Hidden Markov and Maximum Entropy models, Context-Free Grammar, Grammar rules, Treebanks, Normal Forms for grammar, Dependency Grammar, Syntactic Parsing, Ambiguity, Dynamic Programming parsing, Probabilistic CFG, Probabilistic CYK.

UNIT-II Semantics and Pragmatics

Requirements for representation, First-Order Logic, Description Logics: Syntax-Driven Semantic analysis, Semantic attachments Word Senses, Relations between Senses, Thematic Roles, selection restrictions–Word Sense Disambiguation, WSD using Supervised methods.

UNIT-III Discourse Analysis and Lexical Resources

Discourse segmentation, Coherence-Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm-Coreference Resolution-Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet.

UNIT-IV Applications and Techniques

Statistical Alignment and Machine Translation, Text Analysis: Sentiment Analysis, Semantic Similarity, Text-Classification and Clustering, Text Summarization, Topics in Information Retrieval,

Books/ References:

1. ***Daniel Jurafskey and James H. Martin, Speech and Language Processing, Prentice Hall, 2009.***
2. ***Christopher D. Manning and Hinrich Schutze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.***
3. ***Hapke, Hannes, Cole Howard, and Hobson Lane. Natural Language Processing in Action: Understanding, analyzing, and generating text with Python. Simon and Schuster, 2019.***

Text Book

Course Title		**APPLIED MACHINE LEARNING**
Course Number	:	**COE4640**
Credits	:	**4**
Course Category	:	**DE (Departmental Elective)**
Pre-requisite(s)	:	**-**
Contact Hours (L-T-P)	:	**3-1-0**
Type of Course	:	**Theory**

Course Objectives:

The objective of this course is to provide understanding of various machine learning algorithms with their strengths and weaknesses. Furthermore, to give an ability to formulate machine learning problems corresponding to different applications.

Course Outcomes

At the end of this course, students will be able to:

1. Understand and classify different machine learning algorithms and their applications.
2. Analyze neural network models for real-world problems.
3. Formulate different word embedding techniques for text analysis.
4. Design large scale recommendation systems.

SYLLABUS:

UNIT 1: INTRODUCTION

Introduction to Machine Learning, Applications, Recent Developments, Classification of Machine Learning Algorithms, Linear Regression, Logistic Regression, Gradient Descent (stochastic GD, batch GD, mini-batch GD, GD with momentum, RMSProp, ADAM)

UNIT 2: NEURAL NETWORKS

Real world use cases, Activation Functions (Sigmoid, Tanh, ReLU, leaky ReLU, softmax), MLP, CNN, LSTM, GRU, Attention, Model Evaluation (MSE and F-score), Bias Vs Variance, Hyperparameter tuning, Quantization, Knowledge Distillation

UNIT 3: EMBEDDINGS

Real world use cases, Word2Vec, Glove, Sentence Embedding, Image Embedding, Semantic Networks, Cosine Similarity, Location Embedding, User Embedding, Post Embedding, Author Embedding, Property Embedding (Zillow)

UNIT 4: LARGE SCALE RECOMMENDATION ENGINES

Real world use cases, Data Collection – impressions and engagement, System design, Multi-stage ranking pipeline, Recall Stage (VSS, Inverted Index, Relationship Graph Traversal), Point-wise Ranker, Pair-wise Ranker, Policy Violation Identifier, Mixer (Manual Rules Override, Soft-sort)

Books:

1. **Machine Learning: A Probabilistic Perspective. Kevin Murphy. MIT Press, 2012.**
2. **Pattern Recognition and Machine Learning. Christopher Bishop. First Edition, Springer, 2006.**
3. **Goodfellow, Ian: Deep Learning-Ian Goodfellow, Yoshua Bengio, Aaron Courville- Google Books, 2016.**
4. **Géron, Aurélien. Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow. " O'Reilly Media, Inc., 2022.**

Course Title		**DATA MINING**
Course Number	:	**COE4630**
Credits	:	**4**
Course Category	:	**DE (Departmental Elective)**
Pre-requisite(s)	:	**-**
Contact Hours (L-T-P)	:	**3-1-0**
Type of Course	:	**Theory**

Course Objectives

To develop an in-depth understanding of the fundamentals of data mining including basic algorithms like data preprocessing, association rules, classification and Clustering in order to mine patterns in datasets.

Course Outcomes

On completion of this course, the student will be able to:

1. Understand various Data preprocessing methods and their importance for data mining.
2. Understand association Rule Mining and different steps of discovering association rules.
3. Describe and understand various classifiers for data mining.
4. Describe and understand different clustering techniques.

SYLLABUS

UNIT-I DATA MINING

Definition, Data Mining Tasks, Types of Data, Data Quality, Data Preprocessing, Measures of Similarity and Dissimilarity, Data Mining Challenges and Applications.

UNIT-II ASSOCIATION ANALYSIS

Basic Concepts and methods, Frequent Itemset Generation, Apriori Principle, Rule Generation, Maximal Frequent itemsets, FP Growth Algorithm, Evaluation of Association Patterns; Concept Hierarchy.

UNIT-III CLASSIFICATION

Basic Concepts and Techniques, Decision Tree Classifier, Model Overfitting, Model Selection, Model Evaluation; Rule-Based Classifier, Nearest Neighbor Classifier, Logistic Regression, Artificial Neural Network (ANN): Perceptron, Multi-layer Neural Network.

UNIT-IV CLUSTER ANALYSIS

Basic Concepts, Types of clusters and clustering, K-means: Strengths and weaknesses, K-means++, Agglomerative Hierarchical Clustering, DBSCAN, Cluster Evaluation; Recent Trends in Data Mining.

Books:

1. *** Pang-Ning Tan, Michael Steinbach, Anuj Karpatne and Vipin Kumar. Introduction to Data Mining 2nd ed. Pearson Education.**
2. **Han, Jiawei, Micheline Kamber, and Jian Pei. *Data Mining: Concepts and Techniques,* 3rd ed. Morgan Kaufmann Publishers, 2012.**
3. **Ian H. Witten, Eibe Frank, Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Technique, 3rd ed.*, Morgan Kaufmann, 2011**

***Text Book**

Course Outcomes	Program Outcomes (POs)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1									
2	2	2	2	2								1
3	3	2	3	2	3							1
4	3	2	3	2	3							1

Course Title INTERNET TOOLS

Course Number : COE4310/CO431
Credits : 4
Course Category : DE
Pre-requisites :
Contact Hours (L-T-P) : 3-1-0
Type of Course : Theory

Course Outcome

1. Explain the fundamental concepts of Cryptography and basic Network protocols.
2. Demonstrate the working of Proxy servers, DNS, torrents, Firewalls and debugging tools.
3. Explain the working of popular encryption algorithms, digital signatures, certificates and key management.
4. Apply encryption methods to ensure message authentication, message integrity and confidentiality.

Syllabus

UNIT I BASIC NETWORK CONCEPTS
Layers of network, Network layer interconnections, Internet Standard, Ports, Proxy servers, Firewall, NAT/PAT, URIs and relative URL, Domain Name System: DNS lookup, records, MX records etc., finger, whois, Content Delivery.

UNIT II PROTOCOLS AND WWW
TCP/IP, UDP, USENET, Internet Control (ICMP), Debugging tools like ping and tracert, IGMP, Internet Management (SNMP). WWW: HTTP, Static and Dynamic web pages with example of JavaScript and php. Markup Languages. Wireshark introduction.

UNIT III ELEMENTARY SOCKET PROGRAMMING
Client/Server model of Interaction, TCP Sockets, TCP Client-Server, UDP Sockets, Socket Options, UNIX System Calls

UNIT IV INERNET SECURITY
Introduction to Cryptography, Encryption Techniques and Authentication Protocols, Digital Signature, Management of Keys. Components of Firewall. Introduction to SSL.

Books and reference material:
1. Forouzan, “TCP/IP Protocol Suit”, 4th ed, Tata McGraw Hill, 2010.
2. Douglas E. Comer: “’INTERNETWORKING WITH TCP/IP (Volume 1)”, 4th ed, Prentice Hall.
3. A. S. Tanenbaum: “Computer Network”, 5th ed, Prentice Hall.
4. W. R. Stevens: “Unix Network Programming”, Prentice Hall.
5. Behrouz Forouzan, “Cryptography & Network Security”, McGraw Hill, 2008.
6. William Stallings : “Cryptography and Network Security Principles and Practices”, 4th Ed, Prentice Hall, 2005.

Course Outcomes	Program Outcomes (POs)										
	a	b	c	d	e	f	g	h	i	j	k
1	x				x	x			x		
2	x	x			x						x
3								x			
4						x			x		x